# Master's Thesis

# AI-Based Malaria Diagnostic Method Using Resolution-Limited Microscopic Thick Blood Smear Images


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# Abstract

Malaria detection strongly depends on microscopy and trained personnel, which can limit fast diagnosis and treatment, especially in resource-limited regions where access to such tools may be limited. This work focuses on developing a simpler deep learning approach for malaria prescreening from thick blood smear images, with future deployment using a combination of microscope, ocular adapter and smartphone.

To achieve this goal, the NLM/NIH dataset was used to train and evaluate multiple models based on ResNet50, ResNet18 and MobileNetV3-Small. The task was to classify complete images as either infected or uninfected. Several simulated resolution levels were tested, starting from high resolution at 224 $\times$ 224 pixels down to the lowest resolution of 56 $\times$ 56 pixels. Grad-CAM and Grad-CAM++ were used to analyse which image regions influenced the model predictions.

All architectures showed very high performance at high resolution. The highest accuracy of 99.69% was achieved by both ResNet18 and MobileNetV3-Small, while ResNet50 achieved 99.58%. At the lowest reduced resolution, the main decrease was observed in specificity. After a two-stage optimisation, MobileNetV3-Small achieved an accuracy of 92.76%, a sensitivity of 96.40%, a specificity of 79.31% and an F1-score of 95.45%. Grad-CAM and Grad-CAM++ showed that the models sometimes focused on image borders and dark background areas. To investigate this behaviour further, an external test was performed using the AIR Uganda dataset, which has different image characteristics. ResNet50 and MobileNetV3-Small still detected most infected images at LR3, while ResNet18 showed lower performance.

The achieved results show that MobileNetV3-Small is a promising lightweight model for future deployment. The developed approach is still at the research prototype stage and requires further evaluation before real-world deployment. The next important step is to evaluate the models using real images captured with the microscope, developed ocular adapter and smartphone setup under realistic deployment conditions.

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# 1 Introduction

To this day, malaria continues to pose a major global health challenge. The World Health Organisation (WHO) reports that in 2020, about 241 million people were infected with malaria worldwide, and approximately 627 000 lives were lost to the disease[2]. Among all regions, Africa bears by far the heaviest burden, accounting for the vast majority of cases and deaths[2]. Early diagnosis is especially important because malaria can develop into a severe and potentially life-threatening disease if treatment is delayed. Prompt treatment within 24 hours after the onset of fever is recommended and plays an important role in reducing malaria-related mortality, particularly among children under five years of age[1]. Although malaria vaccines are available, vaccination alone does not provide complete protection and therefore does not replace the need for reliable diagnosis and treatment[1]. Considering these limitations, this work addresses the need for more efficient diagnostic approaches.

Although microscopic examination of blood smears is still considered the gold standard for malaria diagnosis, it has several limitations. The examination consumes time and requires trained and experienced personnel. The result can also depend on the microscopist, which can lead to differences between observations. This becomes particularly challenging in malaria-endemic regions where trained personnel and laboratory resources may be limited [1, 3]. Other diagnostic methods, such as rapid diagnostic tests (RDTs) and polymerase chain reaction (PCR), are available, but they also have limitations and therefore do not completely replace microscopy [1]. These limitations have increased the interest in computer-aided and automated analysis of blood smear images. Earlier approaches often relied on manually designed image features, which can depend strongly on the dataset and imaging conditions. Deep learning, especially convolutional neural networks (CNNs), provides another approach by learning relevant features directly from image data instead of defining them manually [3, 4]. This makes CNNs a promising approach for supporting and automating malaria diagnosis. In addition, automated image analysis can be combined with smartphone-based image acquisition, which offers a practical and relatively low-cost solution for settings with limited resources [1]. This idea also fits the overall approach of this project, where a smartphone and ocular adapter are used to digitise microscope images for further AI-based analysis.

This thesis is part of a three-part Master's project at the University of Applied Sciences and Arts Hildesheim/Holzminden/Göttingen (HAWK) that focuses on the development of a practical AI-based approach for malaria diagnosis from Giemsa-stained thick blood smears. The project is conducted in cooperation with the Ethiopian Artificial Intelligence Institute (EAII) in Addis Ababa, Ethiopia, and CarByte Engineering GmbH in Germany. CarByte provides supervision during the development of the three individual projects. The AI algorithms are developed in Python so that they can also be used and further developed by EAII. The overall project is divided into three connected Master's projects. Nabil Laaroussi focuses on the development of an ocular adapter that allows microscope images to be captured using a smartphone. Zaineb Lerhmame works on a resolution-adaptive simulation database to study how reduced image resolution affects the diagnostic performance. This thesis focuses on the development and evaluation of the AI-based diagnostic method, with particular attention to its performance under reduced image resolution relevant to the future ocular-adapter setup. Together, these parts contribute to a prototype in which images from an analog microscope can be digitised using a smartphone and then processed by a PC-based diagnostic tool. The current project focuses on the development of a first prototype. Further improvements, especially regarding the robustness required for field use, can be addressed in future work.

The objective of this thesis is to develop and evaluate a deep learning-based method for automatic malaria prescreening from Giemsa-stained thick blood smear images. A particular focus is placed on the influence of reduced image resolution, since the diagnostic method should also work with the image quality provided by the smartphone-based ocular adapter. The developed approach is evaluated at different image resolutions and benchmarked against existing approaches to assess its suitability for the overall diagnostic system.

To achieve the objective of this thesis, the work is divided into several main steps. First, the existing literature on AI-based malaria diagnosis is reviewed and the relevant fundamentals of deep learning are presented. The next step focuses on collecting suitable image data from available malaria databases and preparing the data for further processing. Based on the findings from the literature, suitable neural network architectures are then selected, implemented, tested, and optimised. This includes investigating the network architecture and relevant hyperparameters. Finally, the developed approach is benchmarked against existing methods from the literature, followed by an evaluation and documentation of the results.

This thesis focuses on the development and evaluation of the AI-based diagnostic method and does not cover the development of the hardware used for image acquisition. The design and construction of the smartphone-based ocular adapter are part of a separate thesis by Nabil Laaroussi. Similarly, the development of the resolution-adaptive simulation database is carried out by Zaineb Lerhmame and is not part of this work. The diagnostic task in this thesis is limited to the binary classification of Giemsa-stained thick blood smear images as infected or uninfected. Parasite localisation, quantification of parasitemia, species-level classification, and the analysis of thin blood smears are not covered. Furthermore, clinical validation and field testing in Ethiopian health centres are outside the scope of this thesis.

# 2 System Overview

As described in Introduction, this thesis is part of a larger system for AI-based malaria diagnosis from microscopic blood smear images. This chapter presents the overall system architecture and explains how its main components work together.

## 2.1 System Architecture

The overall system combines image acquisition, image processing, and AI-based diagnosis. Figure 2.1 gives an overview of the complete diagnostic process and shows how the different parts of the project are connected.

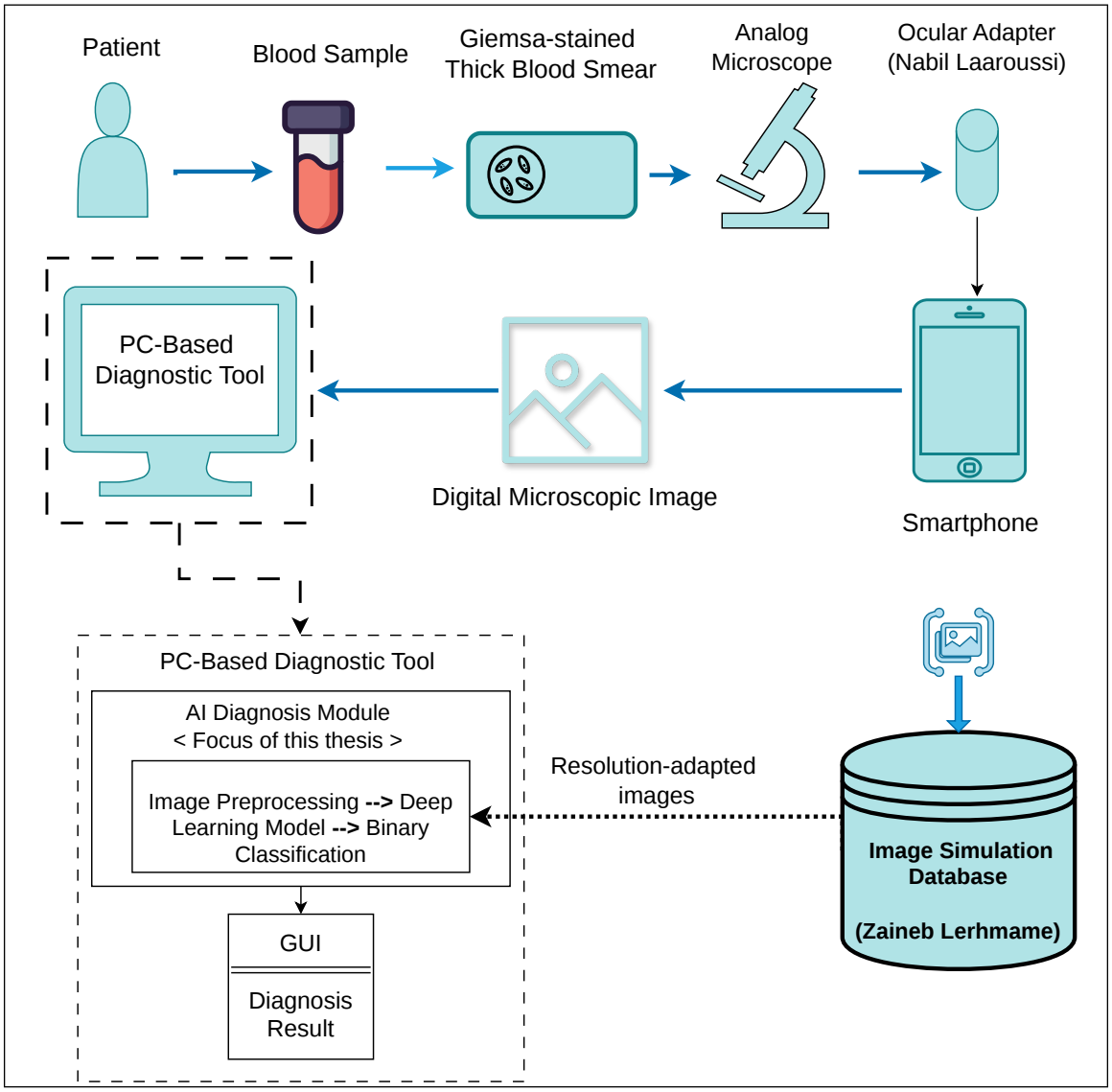



Figure 2.1: Overall architecture of the malaria diagnosis system.

The diagnostic process starts with a blood sample from the patient. The sample is prepared as a Giemsa-stained thick blood smear and examined with an analog microscope. The ocular

adapter, developed by Nabil Laaroussi as part of his Master's project, connects a smartphone to the microscope. The smartphone camera is then used to capture a digital image of the blood smear.

The captured image is transferred to a PC-based diagnostic tool. The AI diagnosis module processes the image and classifies it as infected or uninfected. This part is the main focus of this thesis. The classification result is then shown to the user through a graphical user interface (GUI).

The overall system also includes the image simulation database developed by Zaineb Lerhmame as part of her Master's project. It provides images at different resolutions that can be used with the AI diagnosis module. Since this database supports the development and evaluation of the diagnostic method, it is shown separately from the main diagnostic process in the system architecture.

### 2.1.1 Ocular Adapter

The ocular adapter is used to connect a smartphone to the eyepiece of an analog microscope. Its main purpose is to allow the smartphone camera to capture the microscopic image of the Giemsa-stained thick blood smear. In this way, the microscope image can be digitized and made available for further processing by the PC-based diagnostic tool. The adapter therefore forms the connection between the microscopic examination and the digital image processing part of the system. The detailed design and development of the ocular adapter are part of a separate Master's project by Nabil Laaroussi and are therefore not covered in this work.

### 2.1.2 Image Simulation Database

The image simulation database provides images at different resolutions for the development and evaluation of the AI diagnosis module. Its purpose is to simulate changes in image resolution and make it possible to investigate how these changes affect the diagnostic performance. This is important for the overall system because the image quality provided by the smartphone and ocular adapter may differ from the original microscopic images. The simulation database is used as a supporting component and is not part of the direct diagnostic process from the patient to the final result. The development of the image simulation database is part of a separate Master's project by Zaineb Lerhmame and is therefore not covered in this work.

### 2.1.3 AI Diagnosis Module

The AI diagnosis module is the main focus of this thesis. It receives the digital microscopic images and processes them before they are passed to a deep learning model. The model then performs a binary classification and determines whether an image is infected or uninfected. The classification result is provided to the graphical user interface, where it can be displayed to the user. Since the image resolution can vary within the overall system, the performance of the AI diagnosis module at different image resolutions is also investigated in this work.

### 2.1.4 Deployment Context

The system is intended for use in a health center under controlled indoor conditions. The microscope images are captured using the smartphone and ocular adapter, while the diagnosis is performed using the PC-based diagnostic tool. The AI algorithms are implemented in Python so that they can later be used and further developed by the Ethiopian Artificial Intelligence Institute (EAII).

At this stage, the project focuses on the development of a prototype. A later system for field use may require additional improvements, especially regarding robustness under different operating conditions. These requirements are discussed in more detail in chapter 4.

This chapter presented the overall structure of the malaria diagnosis system and described the role of its main components. The next chapter reviews existing research and approaches related to AI-based malaria diagnosis and provides the background for the development of the diagnostic method used in this thesis.

# 3 State of Research

This chapter provides the background and research basis needed for the development of the diagnostic method. This literature review is a continuation of the work carried out by Kouom Nankam with additional studies that are especially relevant to the focus of this thesis [5]. It first introduces the basic medical principles of malaria diagnosis and the main concepts of deep learning that are relevant for this work. Based on this, existing AI-based approaches for malaria diagnosis are reviewed and compared with regard to their data, model types, evaluation methods, and the actual practical use. This chapter starts with the basic medical background of malaria diagnosis and then introduces the deep learning methods that are relevant for this work. After that, previous research approaches are presented and compared. The chapter ends by identifying the main research gap and explaining how it is related to this thesis.

## 3.1 Literature Basis and Previous Work

The literature review in this thesis is based on previous work carried out at HAWK Göttingen. In February 2026, Kouom Nankam completed a systematic literature review on AI-based malaria diagnosis using microscopic blood images. His research covered publications from 2022 to 2025 from IEEE Xplore, ScienceDirect, SpringerLink, and PubMed. His search used combinations of terms such as ”malaria diagnosis”, ”deep learning”, ”convolutional neural networks”, ”vision transformer”, and ”microscopic blood smear”. His analysis first began with reviewing the titles and abstracts to check their relevance. After that, the selected studies were read and analysed in full [5].

The review contained 38 references and focused on datasets, preprocessing methods, model architectures, training and validation strategies, evaluation metrics, explainability, and practical applicability. The work was literature-based and did not include its own model implementation or experimental evaluation [5].

Although the review covered different AI approaches for malaria diagnosis, it focused mainly on thin blood smear images. Kouom explicitly excluded thick blood smear analysis from the scope of the work. He pointed out that thick blood smears provide higher sensitivity for parasite detection, but that their complex visual structure and high cell density make automated image analysis and reliable annotation more difficult [5]. This limitation is relevant for the present thesis since the diagnostic method developed here focuses specifically on thick blood smear images.

To extend the previous literature review, an additional search was carried out for studies that were directly relevant to the research questions of this thesis. Search terms included combinations of ”malaria”, ”thick blood smear”, ”deep learning”, ”MobileNet”, ”object detection”, ”image resolution”, ”smartphone microscopy”, and ”explainable AI”. The identified studies were reviewed with focus on the parts relevant to this thesis, especially the methods, tools, datasets, and reported results. The findings were then compared with the previous HAWK literature review to avoid unnecessary duplication. Particular attention was given to studies involving thick blood smear images, patient-wise evaluation, reduced image resolution, and explainability.

Table 3.1: Comparison of the HAWK literature review and the literature review of this thesis.

| **Aspect** | **Kouom Nankam** | **This thesis** |
|---|---|---|
| Literature scope | Publications from 2022–2025 | Previous review extended with additional relevant studies |
| Blood smear type | Mainly thin blood smears. Thick-smear analysis excluded from the review scope | Mainly thick blood smears, thin blood smears included only in the literature review |
| Main focus | General state of AI-based malaria diagnosis | AI-based diagnosis with focus on classification, image resolution, efficiency, and explainability |
| Studies used | 38 references | 35 references including the work of Kouom Nankam |
| Experimental work | No implementation or experiments | Model implementation, training, testing, and benchmarking |

## 3.2 Malaria and Microscopic Diagnosis

Malaria is an infectious disease caused by parasites from the genus Plasmodium. Five species are known to cause malaria in humans: P. falciparum, P. vivax, P. ovale, P. malariae, and P. knowlesi. Among them, P. falciparum is the most dangerous species and is responsible for most deaths caused by severe malaria [1]. During the blood stage of the infection, different parasite stages can be seen under the microscope. These include immature trophozoites, also known as ring forms, mature trophozoites, schizonts, and gametocytes [1]. These different stages are important for the microscopic diagnosis of malaria.

Giemsa staining is commonly used to make malaria parasites visible in blood smears. The WHO recommends Giemsa staining for both thick and thin blood films [6]. Before staining, the thin blood film is fixed with methanol. The thick blood film should not come into contact with methanol because the red blood cells need to undergo haemolysis. After staining, the slides are rinsed with buffered water and left to dry before they are examined under the microscope [6].

Two main types of blood smears are used for microscopic malaria diagnosis: thick and thin blood smears. Thick smears are mainly used to check whether malaria parasites are present, while thin smears preserve the structure of the red blood cells and are therefore more suitable for identifying the Plasmodium species [1].

Figure 3.1 shows the visual difference between a thick and a thin Giemsa-stained blood smear.

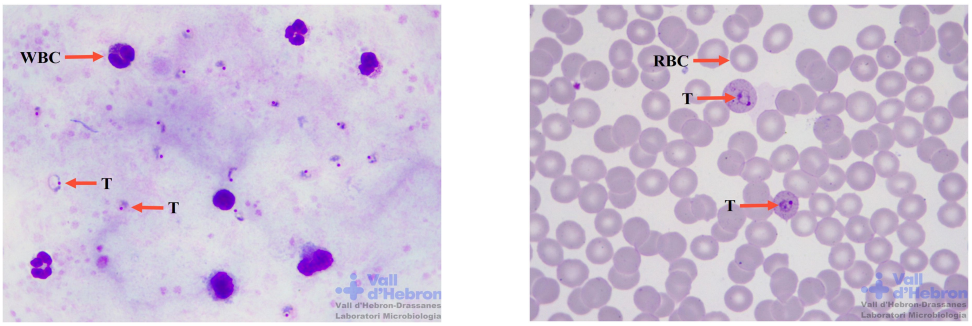



Figure 3.1: Comparison of Giemsa-stained blood smears. The left image shows a thick blood smear and the right image shows a thin blood smear. Source: Maturana et al. [1].

In the figure, WBC stands for white blood cells, RBC for red blood cells, and T for trophozoites.

Thick blood smears are more sensitive than thin blood smears and can therefore detect lower parasite levels. Maturana et al. report that thick blood smears are approximately 20 to 40 times more sensitive than thin blood smears [1].

Thin blood smears, on the other hand, keep the individual red blood cells visible. This makes it possible to examine the parasites together with the infected cells and makes species identification easier. The main differences between thick and thin blood smears are shown in Table 3.2 [1].

Table 3.2: Comparison of thick and thin blood smears for malaria microscopy.

| **Aspect** | **Thick blood smear** | **Thin blood smear** |
|---|---|---|
| Main purpose | Detection and screening of malaria parasites | Identification of the Plasmodium species |
| Sensitivity | Higher, reported to be about 20–40 times more sensitive | Lower compared with thick smears |
| Red blood cells | Broken down during staining | Remain intact |
| Parasite visibility | Parasites are concentrated in a dense smear area | Parasites can be examined inside individual red blood cells |
| Image structure | Dense and less clearly structured | Individual cells can be seen more clearly |

Thick blood smears can also be used to estimate the parasite density, not only to detect an infection. For this, the number of parasites is counted together with the number of white blood cells (WBCs). According to the WHO procedure, these values can then be used to calculate the parasite density [7]:

$$D = \frac{N_P \cdot C_{\mathrm{WBC}}}{N_{\mathrm{WBC}}}, \tag{3.1}$$

where $D$ is the estimated parasite density in parasites/$\mu$L, $N_P$ is the number of parasites counted, $N_{\mathrm{WBC}}$ is the number of white blood cells counted, and $C_{\mathrm{WBC}}$ is the patient's white blood cell concentration per $\mu$L.

If the actual WBC concentration is not available, the WHO recommends using an assumed value of 8000 WBC/$\mu$L [7]. Equation 3.1 then becomes

$$D = \frac{N_P \cdot 8000}{N_{\mathrm{WBC}}}. \tag{3.2}$$

Parasite density estimation is not part of the method developed in this thesis. The equations are only included to show how parasite concentration is calculated during conventional microscopic examination. The task of this work focuses instead on classifying thick blood smear images as infected or uninfected.

Although thick blood smears are very sensitive for parasite detection, their image structure makes automatic analysis more difficult. In contrast to thin smears, individual red blood cells are no longer clearly visible, and parasites appear together with other stained blood components and possible artefacts [8]. Kouom Nankam also describes the complex structure and high cell density of thick smears as challenges for automated image analysis and annotation [5]. For this reason, this thesis focuses specifically on the automatic classification of thick blood smear images. Later experiments also investigate how reduced image resolution affects this task.

## 3.3 Deep Learning Fundamentals

Since this thesis is based on deep learning, this section introduces the main concepts that are relevant for the developed method. It first explains the basic idea of deep learning and then focuses on convolutional neural networks, transfer learning, image analysis tasks, and the evaluation metrics used later in this work.

### 3.3.1 Convolutional Neural Networks

A CNN processes an image through several layers. The first layers usually learn simple image features such as edges, colours, and textures. Deeper layers combine these simple features into more complex patterns that are useful for the final prediction [9]. This hierarchical feature extraction is one of the main reasons why CNNs are suitable for image classification.

The main operation of a CNN is the convolution. A small filter, also called a kernel, moves across the input image or feature map and calculates a new feature map. Following the standard formulation of convolution used in deep learning [9], A simplified two-dimensional convolution operation used in CNNs can be written as:

$$Y(i,j) = \sum_m \sum_n X(i+m, j+n)K(m,n) + b \tag{3.3}$$

where $X$ represents the input, $K$ is the convolution kernel, $b$ is a bias term, and $Y$ is the resulting feature map. During training, the values of the kernels are learned automatically from the data.

After a convolution, an activation function is normally applied. A commonly used activation function is the Rectified Linear Unit (ReLU) [9], which is defined as:

$$\mathrm{ReLU}(x) = \max(0, x) \tag{3.4}$$

ReLU keeps positive values while setting negative values to zero. This introduces non-linearity and allows the network to learn more complex relationships between the input image and the target class [9].

**Type and extent of AI use**
Claude was used to extract useful information from the book by Goodfellow et al. The extracted information were used for the deep learning fundamentals section.

**The individual contribution**
The extracted information were selected based on what was needed for this section and was used according to the structure of the section. The information was then used to explain the main deep learning fundamentals that are relevant to this thesis.

**Critical evaluation of the AI contribution**
The book by Goodfellow et al. contains more than 800 pages. By using Claude, the relevant information could be found and extracted more efficiently. This also saved time that could be used to analyse other references that are more directly related to the topic of this thesis.

The development of CNN architectures has resulted in networks with different levels of complexity. AlexNet demonstrated the potential of deep CNNs for large-scale image classification [10]. VGG later increased the network depth using repeated small convolution filters [11]. ResNet introduced

residual connections that allow information to bypass individual layers and make the training of deeper networks easier [12]. Other architectures focus more strongly on computational efficiency. MobileNet uses depthwise separable convolutions to reduce computational cost [13], while SqueezeNet was designed to strongly reduce the number of model parameters [14]. These differences between CNN architectures are relevant when selecting a model based not only on classification performance but also on computational requirements.

CNNs are not the only approach used in modern image analysis. Vision Transformers (ViTs) apply the Transformer concept to images by dividing an image into smaller patches and processing the relationships between these patches using attention mechanisms [15].

### 3.3.2 Transfer Learning

Training a deep neural network completely from scratch generally requires a large amount of labelled data. Transfer learning provides an alternative by reusing knowledge learned from another task. In transfer learning, knowledge from a previously learned task is transferred to support learning on a new task [16].

For image classification, pretrained models can be used as a starting point for a new task. The learned features of these models can then be reused for another image classification problem [4].

When pretrained models are further trained using a new dataset, this process is commonly referred to as fine-tuning. This approach is also used in malaria image analysis, for example by Maturana et al., who fine-tuned pretrained detection models using their thick blood smear dataset [17].

During training, the model parameters are updated by minimising a loss function. For classification tasks, cross-entropy loss is commonly used to measure the difference between the predicted class probabilities and the true class label [9]. For a classification problem with $C$ classes, the cross-entropy loss can be written as:

$$L = -\sum_{c=1}^{C} y_c \log(p_c) \tag{3.5}$$

where $y_c$ represents the true class label and $p_c$ the predicted probability for class $c$.

During model development, the available data are normally separated into training, validation, and test data. The training set is used to update the model parameters. The validation set is used to observe the model during development and to support decisions such as model or hyperparameter selection. The test set should remain separate and is used for the final evaluation of the trained model. This separation is important because evaluating the model on data that were already used during training can lead to an unrealistic estimation of its performance [9].

**Type and extent of AI use**
Claude was used to extract useful information from the book by Goodfellow et al. The extracted information was used for the deep learning fundamentals section.

**The individual contribution**
The extracted information was selected based on what was needed for this section and was arranged according to the structure of the chapter. It was then used to explain the main deep learning fundamentals that are relevant to this thesis.

**Critical evaluation of the AI contribution**
The book by Goodfellow et al. contains more than 800 pages. By using Claude, the relevant information could be found and extracted more efficiently. This also saved time that could be used to analyse other references that are more directly related to the topic of this thesis.

### 3.3.3 Image Analysis Tasks

The studies reviewed in this thesis use different image analysis tasks. The main tasks relevant to this work are classification, object detection, and segmentation.

Image classification assigns a class to an image or image region. For example, a binary classifier can distinguish between infected and uninfected samples [4]. However, classification does not show where an object is located in the image.

Object detection also identifies the position of objects, usually using bounding boxes. YOLO is a common example of an object detection approach [18]. In malaria analysis, this can be used to locate individual parasites or blood cells.

Image segmentation works at pixel level and assigns a label to individual pixels [19]. This allows structures such as cells or background regions to be separated more precisely.

Since these tasks solve different problems, their results and evaluation metrics should not be compared directly without considering the task being performed.

### 3.3.4 Evaluation Metrics

The performance of a binary classification model can be described using true positives ($TP$), true negatives ($TN$), false positives ($FP$), and false negatives ($FN$). In this thesis, infected samples are treated as the positive class and uninfected samples as the negative class. The following evaluation metrics are based on these values [20]:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{3.6}$$

Sensitivity, also called recall, describes how many infected samples are correctly identified:

$$\text{Sensitivity} = \text{Recall} = \frac{TP}{TP + FN}. \tag{3.7}$$

Specificity describes how many uninfected samples are correctly identified:

$$\text{Specificity} = \frac{TN}{TN + FP}. \tag{3.8}$$

Precision describes how many samples predicted as infected are actually infected:

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{3.9}$$

The F1-score combines precision and recall:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{3.10}$$

These metrics describe different parts of the model performance. Sensitivity is especially important for showing how reliably infected samples are detected, while specificity shows how reliably uninfected samples are identified [20].

The Receiver Operating Characteristic (ROC) curve shows the relationship between the true-positive rate and false-positive rate at different classification thresholds. The Area Under the ROC Curve (ROC-AUC) summarises this behaviour in one value. Delahunt et al. point out that metrics such as ROC-AUC and F1 should be interpreted carefully in malaria diagnosis and emphasise the importance of patient-level evaluation [20].

Object detection studies can also report Average Precision (AP) or mean Average Precision (mAP). These metrics are relevant to the object detection studies discussed later in this chapter, but they are not used to evaluate the classifiers developed in this thesis.

## 3.4 AI-Based Malaria Diagnosis from Microscopic Images

Automated malaria diagnosis has developed from traditional image-processing methods towards deep learning-based approaches. Earlier methods often used manually designed image features, while more recent approaches use convolutional neural networks to learn relevant features directly from microscopic images [1, 21].

The literature includes different tasks such as image classification, parasite detection, species identification, and patient-level diagnosis. Kassim et al., for example, combined parasite detection and classification to distinguish between P. falciparum, P. vivax, and uninfected samples at image and patient level [22].

Both thin and thick blood smear images have been used for automated malaria analysis. Thick blood smears are especially relevant for parasite screening, but their dense image structure makes automatic analysis more difficult [1]. For this reason, the following sections focus mainly on approaches using thick blood smear images.

Smartphone-based microscopy has also become relevant for automated malaria diagnosis. Quinn et al. used CNNs for malaria detection from thick blood smear images captured using smartphones [21]. Such approaches are relevant to this thesis because the future system is also intended to combine a microscope, smartphone, and automated image analysis.

## 3.5 Thick Blood Smear Datasets and Image Acquisition

For the practical work of this thesis, three thick blood smear datasets were considered: the datasets provided by the National Library of Medicine (NLM), the Uganda dataset presented by Quinn et al., and the Drassanes–Vall d'Hebron dataset.

The NLM provides thick blood smear images from patients infected with P. falciparum and P. vivax, as well as images from uninfected patients. Yang et al. published a dataset containing 1819 images from 150 patients infected with P. falciparum. The images were captured using a smartphone camera connected to a microscope and contain 84 961 manually annotated parasites [8]. Kassim et al. later used a combined NLM collection containing 5972 images from 350 patients with P. falciparum, P. vivax, and uninfected samples [22].

The Uganda dataset presented by Quinn et al. contains 1182 thick blood smear images with 7245 annotated parasites [21]. The images were captured using a smartphone microscope setup. Since the dataset contains malaria-positive images and parasite-level annotations, it is mainly suitable for parasite detection rather than infected-versus-uninfected classification. In this thesis, it was later used as an external test dataset.

The Drassanes–Vall d'Hebron dataset contains 2571 labelled thick blood smear images [23]. The images were captured using either a Leica microscope camera or a Samsung Galaxy S20 connected to the microscope. The dataset contains different Plasmodium species and object-level annotations. However, all included thick blood smear samples are malaria-positive, which means that the dataset cannot directly provide an uninfected class for the binary classification task used in this thesis.

Table 3.3 gives an overview of the three datasets.

Table 3.3: Overview of the thick blood smear datasets considered in this thesis.

| **Dataset** | **Size** | **Acquisition** | **Main characteristics** |
|---|---|---|---|
| NLM datasets | 5972 images / 350 patients reported by Kassim et al. | Smartphone microscopy | P. falciparum, P. vivax, and uninfected samples [8, 22] |
| Uganda / AIR | 1182 images / 7245 annotated parasites | Smartphone-microscope setup | Malaria-positive thick smear images with parasite-level annotations [21] |
| Drassanes–Vall d'Hebron | 2571 labelled images | Leica microscope camera or Samsung Galaxy S20 | Multiple Plasmodium species and object-level annotations; malaria-positive samples [23] |

The NLM datasets were selected for the main training and internal evaluation because they contain both infected and uninfected patients. The P. falciparum and P. vivax images could therefore be combined into one infected class and compared with the uninfected class.

The high original image resolution was also useful for the resolution experiments in this thesis because lower-resolution versions could be generated from the same images. The final dataset preparation, class definition, patient-wise splitting, and preprocessing are described in Chapter 6.

## 3.6 CNN-Based Classification Approaches

CNNs are widely used for automated malaria image classification because they can learn relevant image features directly from the training data. Different approaches have used pretrained CNNs, custom CNN architectures, or a combination of detection and classification.

Rajaraman et al. investigated pretrained CNNs for the classification of parasitized and uninfected cells in thin blood smear images [4]. The evaluated architectures included AlexNet, VGG-16,

ResNet-50, Xception, and DenseNet-121. The study showed that features learned by pretrained CNNs can be reused for malaria image classification.

For thick blood smears, Yang et al. developed a method that combines Iterative Global Minimum Screening (IGMS) with a custom CNN [8]. IGMS first selects possible parasite candidates from the microscopic image. The CNN then classifies the extracted patches as parasite or background. Their CNN achieved an accuracy of 93.46%, a sensitivity of 92.59%, and a specificity of 94.33%. The study is particularly relevant because it uses smartphone-acquired thick blood smear images. However, the classification is performed on selected image patches instead of the complete microscopic image.

Kassim et al. also combined detection and classification for thick blood smear images [22]. Their PlasmodiumVF-Net first detects parasite candidates using Mask R-CNN and then uses ResNet50 to reduce false-positive detections. The final system provides decisions at image and patient level and distinguishes between P. falciparum, P. vivax, and uninfected samples. This study is especially relevant because it uses the same NLM dataset family that is used in this thesis.

These studies show that CNNs can be used in different ways for malaria analysis. Rajaraman et al. used pretrained CNNs for cell classification, while Yang et al. and Kassim et al. combined candidate detection with CNN-based classification. In contrast, this thesis investigates direct binary classification of complete thick blood smear images without a separate parasite detection step.

ResNet50 and ResNet18 are used as the main residual architectures in this thesis. Lightweight architectures such as MobileNet are discussed separately in Section 3.8.

## 3.7 Object Detection Approaches

Object detection is used when individual parasites or blood cells need to be located inside a microscopic image. Unlike image classification, the model does not only predict a class but also identifies the position of the detected objects. This is mostly useful in malaria analysis because several parasites can appear within the same microscopic field.

Maturana et al. investigated object detection on thick blood smear images as part of the iMAGING system [17]. They compared YOLOv5x, Faster R-CNN, SSD, and RetinaNet using a dataset of 2571 labelled images. The models detected leukocytes, early trophozoites, and mature trophozoites. YOLOv5x achieved a precision of 92.10%, a recall of 93.50%, an F-score of 92.79%, and an mAP@0.5 of 94.40%. This study is particularly relevant because it combines thick blood smear analysis with smartphone-based image acquisition and automated microscopy.

Vásquez Ascate et al. developed a smartphone-based object detection system for thick blood smears [24]. Their dataset contained 8625 images from 400 slides captured using three different smartphones. Several lightweight object detection models were evaluated. YOLOv11n achieved an accuracy of 90.92% and an F1-score of 93.8%. The study is relevant to this thesis because it combines thick blood smears, smartphone image acquisition, and models designed for mobile deployment.

These studies show that object detection can be used to locate individual parasites inside thick blood smear images. However, this requires object-level annotations and a different processing approach than the method used in this thesis. The present work instead focuses on binary classification of the complete image as infected or uninfected without directly locating individual parasites.

## 3.8 Lightweight Networks and Computational Efficiency

For this thesis, classification performance is not the only important point. The computational requirements of a model are also important because the final system is intended for deployment with limited resources. A smaller model can be more suitable for limited hardware even if its accuracy is slightly lower than that of a larger model.

The MobileNet family was developed for efficient image processing on mobile and embedded devices [13]. MobileNet reduces the computational cost by using depthwise separable convolutions instead of standard convolutions. This makes the architecture smaller and more efficient while still allowing useful image features to be learned.

Lightweight models are also relevant in recent malaria research. Vásquez Ascate et al. evaluated several lightweight object detection models for smartphone-based analysis of thick blood smear images [24]. Their work shows the importance of model size and computational efficiency when malaria analysis is intended for mobile devices.

In this thesis, MobileNetV3-Small is used as the lightweight alternative to ResNet50 and ResNet18. The objective is to investigate whether a smaller model can still achieve good classification performance, especially when the image resolution is reduced. This is relevant for the future system because the model should not only perform well, but should also remain suitable for practical deployment.

## 3.9 Influence of Image Resolution

Image resolution is important in this thesis because malaria parasites can appear as small structures inside a microscopic image. When the resolution is reduced, some visual details can be lost, which may affect the performance of the model.

In the studies reviewed for this thesis, image resolution is usually defined by the dataset or the image acquisition setup. Yang et al., for example, used smartphone-acquired thick blood smear images with an original resolution of $3024 \times 4032$ pixels [8]. Their method detects possible parasite regions and then classifies smaller image patches. However, different resolution levels were not compared systematically.

Kassim et al. also used high-resolution smartphone images from the NLM thick blood smear datasets [22]. Their work focuses on parasite detection and malaria classification, but does not investigate how the model performance changes when the image resolution is reduced.

This leaves an open question about how much image detail can be reduced before the classification performance begins to decrease. This is relevant for the system developed in this project because the final images will be captured using a microscope, ocular adapter, and smartphone, and their image characteristics may differ from the original dataset images.

For this reason, this thesis investigates several controlled resolution levels using the same thick blood smear images. This makes it possible to study how the loss of image detail affects whole-image classification while keeping the underlying image content and patient data unchanged.

## 3.10 Explainability in Malaria Diagnosis

Deep learning models can achieve high classification performance, but it is often difficult to understand which parts of an image influenced the prediction. In medical image analysis, this

is important because the model should focus on relevant image regions instead of unrelated structures or background information.

Rajaraman et al. investigated this problem for malaria classification using thin blood smear images [25]. They used different visualisation methods, including saliency maps, occlusion sensitivity, class activation maps, and Grad-CAM. Their results showed that the trained CNNs focused on regions containing malaria parasites.

Grad-CAM uses the gradients of a predicted class to create a heatmap showing which regions of the image contributed most to the prediction [26]. In this thesis, Grad-CAM and Grad-CAM++ are used to inspect true-positive, true-negative, false-positive, and false-negative predictions.

This helps to investigate whether the models focus on relevant regions inside the microscopic field or on unrelated areas such as image borders or background.

## 3.11 Comparison of Relevant Approaches

The studies reviewed in the previous sections use different datasets, tasks, and evaluation methods. Their reported results therefore cannot be compared directly. Table 3.4 gives a short overview of the thick blood smear studies that are most relevant to this thesis.

Table 3.4: Comparison of selected thick blood smear approaches.

| **Study** | **Data** | **Task** | **Approach** | **Reported Results** |
|---|---|---|---|---|
| Yang et al. [8] | 1819 images from 150 patients | Parasite candidate classification | Candidate screening + custom CNN | 93.46% accuracy, 92.59% sensitivity, 94.33% specificity |
| Kassim et al. [22] | 5972 images from 350 patients | Parasite detection and image/patient classification | Mask R-CNN + ResNet50 | 83.9% image-level accuracy, 92.3% patient-level accuracy |
| Maturana et al. [17] | 2571 labelled images | Parasite and leukocyte detection | YOLOv5x, Faster R-CNN, SSD, RetinaNet | YOLOv5x: 92.79% F-score, 94.40% mAP@0.5 |
| Vásquez Ascate et al. [24] | 8625 images from 400 slides | Parasite detection | Several lightweight object detection models | YOLOv11n: 90.92% accuracy, 93.8% F1-score |

The comparison shows that the studies use different approaches for thick blood smear analysis. Yang et al. classify selected parasite candidates, while Kassim et al. combine parasite detection with image- and patient-level classification. Maturana et al. and Vásquez Ascate et al. focus mainly on locating parasites inside the image.

The studies also show the importance of smartphone-based image acquisition and models that can be used with limited resources. However, the tasks are different, so their reported performance values should not be compared directly.

Another difference is that the reviewed studies mainly work with the image resolution provided by their dataset or acquisition setup. The effect of systematically reducing the image resolution is not the main focus of these studies.

## 3.12 Research Gap and Relevance to This Thesis

The reviewed studies show that different approaches are used for automated malaria analysis. Yang et al. first select possible parasite regions from thick blood smear images and then classify the extracted patches [8]. Kassim et al. also use a multi-stage approach that combines parasite detection with further classification [22]. In contrast, this thesis investigates binary classification of the complete thick blood smear image as infected or uninfected without locating individual parasites.

Another important point is image resolution. The thick-smear studies of Yang et al. and Kassim et al. use images with a fixed original resolution [8, 22]. The influence of systematically reducing the image resolution is not the main focus of these studies, but it is relevant for the planned system because the images will later be captured using a microscope, ocular adapter, and smartphone, and their image characteristics may differ from the original dataset images.

Explainability is also relevant for this work. Rajaraman et al. showed that Grad-CAM can be used to inspect which regions influence malaria classification [25]. In this thesis, Grad-CAM and Grad-CAM++ are used to investigate whether the models focus on relevant regions inside the microscopic image or on unrelated areas such as image borders or background.

Based on these points, this thesis compares ResNet50, ResNet18, and MobileNetV3-Small using the same dataset and preprocessing conditions. The models are evaluated at different simulated resolution levels to investigate how the loss of image detail affects classification performance. MobileNetV3-Small is also included as a lightweight alternative for future deployment with limited resources.

The main focus of this thesis is therefore the combination of whole-image thick-smear classification, reduced image resolution, different CNN architectures, and explainability. These points form the basis for the method development and experiments described in the following chapters.evaluation, and explainability. These findings provide the basis for the requirements defined in the following chapter.

# 4 Requirements Analysis

This chapter describes the main requirements for the diagnostic system developed in this project. The requirements are based on the project description file provided by Professor A. Ibenthal. These requirements derive from a workshop held on 14 November 2025. In addition to the functional requirements, the chapter describes important technical limitations, dependencies between the three related Master's projects, and the boundaries of this thesis.

## 4.1 Functional Requirements

The requirements workshop defined four main functional requirements for the overall system, which are summarised in Table 4.1.

Table 4.1: Functional requirements of the diagnostic system.

| **Requirement** | **Description** |
|---|---|
| Deployment | The system should be usable in a health center or similar enclosed environment. |
| Smartphone-based image acquisition | Microscopic images should be digitised by a mobile phone attached to an ocular adapter which is connected to an analog microscope. |
| PC-based diagnostic tool | The captured images should be processed and analysed by software running on a computer. |
| Python-based AI algorithm | The AI algorithm should be implemented in Python to support reuse and further development by EAII. |

The first requirement is the deployment environment. The system is intended for use in a health center or similar indoor environment. The project therefore does not target operation in uncontrolled outdoor environments.

The second requirement concerns image acquisition. Existing analog microscopes should be used together with an ocular adapter. A mobile phone is then attached to the adapter and captures digital images through the microscope. This allows existing microscopy equipment to be used without replacing the complete microscope.

The third requirement is a PC-based diagnostic tool. Images captured with the smartphone should be transferred to a computer, where they can be processed and analysed by the diagnostic tool.

The fourth requirement concerns the implementation of the AI algorithm. The algorithm should be developed in Python, so it can later be reused and further developed by the Ethiopian Artificial Intelligence Institute (EAII).

## 4.2 Non-Functional Requirements and Limitations

In addition to the functional requirements, several technical and practical constraints influence the development of the diagnostic system. The main non-functional requirements and limitations are summarised in Table 4.2.

Table 4.2: Non-functional requirements and limitations.

| **NF-Requirement / limitation** | **Description** |
|---|---|
| Resolution compatibility | The AI method should remain suitable for the image resolution achievable by the ocular adapter. The final measured value is still to be provided by the ocular adapter project. |
| Diagnostic performance | Classification performance should be evaluated using suitable metrics. No fixed minimum target is currently defined in the available requirements. |
| Implementation language | The AI algorithms should be implemented in Python. |
| Maintainability | The implementation should be understandable and reusable for later development by EAII. |
| Indoor operating scope | The current system is designed only for controlled indoor use. Additional robustness requirements for field deployment are outside the current project scope. |

An important aspect is the relationship between diagnostic performance and image resolution. The ocular adapter developed within the overall project limits the resolution of the captured images. The AI method should therefore be suitable for the resolution that can realistically be achieved with the adapter.

At the current stage, the final measured resolution of the ocular adapter has not yet been established for this thesis. The corresponding value will be obtained from the adapter development and later included as a concrete system limitation. Until this value is available, the resolution requirement remains open. Similarly, no fixed minimum values for accuracy, sensitivity, or specificity are defined in the available project requirements.

The AI implementation is restricted to Python, as defined in the requirements workshop. This supports code reuse and future development by the Ethiopian Artificial Intelligence Institute. The software should also remain understandable and maintainable so that it can be further developed after completion of the thesis. The project is therefore managed in a GitHub repository, where the code and important development steps are regularly committed and documented.

The current work of the overall project develops and tests the system for a controlled indoor environment. Requirements for a fully portable and robust field system are left for future work.

## 4.3 Dependencies between the Project Components

The diagnostic system is developed through three related Master's projects, which address different parts of the same processing pipeline and therefore depend on each other. Their connection is shown in Figure 4.1.

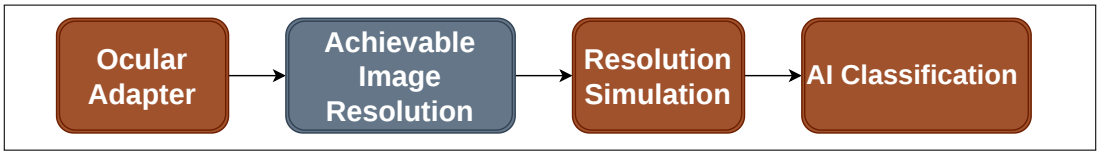



Figure 4.1: Relationship between the three project components.

The first project, carried out by Nabil Laaroussi, focuses on the development of the ocular adapter. The adapter connects the smartphone camera to an analog microscope and determines

the achievable resolution of the image. The resolution is tested using special test images and measured by how many line pairs can still be clearly seen across the image.

This measured resolution represents an important constraint for the AI method developed in this thesis. A classification model may perform well on high-resolution laboratory images but may not maintain the same performance when the resolution is reduced by the image acquisition system. The final adapter resolution therefore has to be considered when evaluating the suitability of the trained models.

The second project, conducted by Zaineb Lerhmame, focuses on the creation of resolution-adaptive simulation data. Existing high-resolution images are processed to simulate lower image resolutions. The approach uses image preprocessing and low-pass filtering to reproduce a defined resolution level. This allows the diagnostic algorithm to be evaluated under image conditions similar to those expected from the ocular adapter.

This thesis represents the AI diagnosis part of the system. It uses malaria microscopy images to develop and compare classification models and investigates their suitability under multiple image resolutions. This work therefore depends on both the acquisition limitations identified by the ocular adapter project and the resolution simulation approach developed in the second project.

## 4.4 Requirements derived from the Research Gap

The literature review in Chapter 3 identified two aspects that are especially relevant for the requirements of this thesis.

Many existing malaria approaches focus on detecting parasites or analysing small image regions instead of classifying the complete microscopic image. In this thesis, the whole thick blood smear image is used for binary classification. The model therefore classifies each image as either infected or uninfected without detecting individual parasites.

The reviewed studies mainly evaluate their models at a fixed image resolution. The influence of systematically reduced resolution is less clearly investigated. Since the ocular adapter introduces a practical limitation on achievable image resolution, the AI method should be evaluated not only on the original high-resolution images but also under reduced-resolution conditions.

These two aspects lead to the following additional requirements for the experimental work:

- The models should support binary whole-image classification of thick blood smear images.
- The same classification approach should be evaluated at different image resolutions.
- Model performance should be compared using the same data split and evaluation procedure.
- The results should allow an assessment of whether the models remain effective when image resolution is reduced.

The measured resolution of the ocular adapter should be used as the main reference once it becomes available. Since the adapter development is still ongoing, the AI models are also tested at lower resolutions to investigate their behaviour under more demanding conditions.

## 4.5 Scope of the Present Thesis

Based on these requirements, this thesis focuses on the AI-based classification component of the diagnostic system. The main task is to classify thick blood smear images as infected or uninfected.

The work includes the comparison of ResNet50, ResNet18, and MobileNet under comparable conditions and at reduced image resolutions, as well as the use of Grad-CAM to better understand model predictions.

The mechanical design of the ocular adapter itself is not part of this thesis. The development of the resolution-adaptive simulation database is also executed in a separate project. Clinical validation, species-level diagnosis, parasite density estimation, and the development of a fully field-ready Mobile Diagnostic Kit are outside the scope of the present work.

The requirements defined in this chapter provide the framework for the concept development and implementation described in the following chapters.

# 5 Concept Development

This chapter converts the findings from the literature review in Chapter 3 and the requirements defined in Chapter 4 into concrete structural decisions for the AI-based classification system.

This section of the thesis explains the background behind selecting the CNN architectures, the processing pipeline, the training strategy, the hyper-parameter selection, the evaluation setup, and the Grad-CAM analysis.

These decisions establish the basis for the implementation described in the next chapter which is Chapter 6.

## 5.1 Selection of CNN Architectures

Three CNN architectures were selected for this thesis: ResNet50, ResNet18, and MobileNet. The selection was based on two main factors from the literature review: classification performance and computational efficiency.

Since classification performance is an important objective of this work, ResNet50 and ResNet18 were selected as the main residual architectures. As discussed in Section 3.6, Rajaraman et al.

MobileNet was selected as the lightweight architecture for the intended system. As discussed in Section 3.8, lightweight CNN architectures are useful when computational resources are limited, which is relevant to this project. MobileNet reduces computational effort by using depth-wise separable convolutions and is therefore suitable for comparing classification performance with a more efficient network design. This is important for the intended PC-based diagnostic system and also for possible future deployment where computational resources are limited.

The three architectures were therefore selected to represent two different goals. ResNet50 and ResNet18 focus more on classification performance, while MobileNet represents the lightweight side of the comparison. This addresses the two aspects of the research gap identified in Section 3.12: achieving strong classification performance while also considering computational efficiency.

## 5.2 Processing Pipeline

The processing pipeline describes how a thick blood smear image is transformed from the original input into a classification result. It represents the AI-related part of the overall system introduced in Chapter 2.

The overall processing pipeline is shown in Figure 5.1.

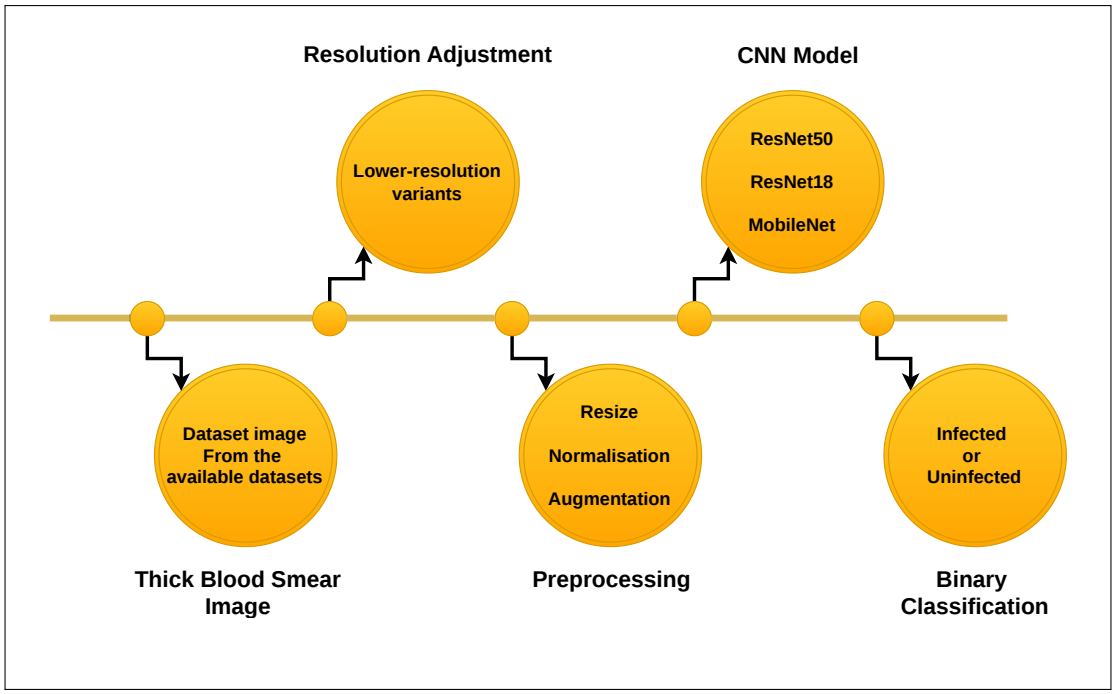



Figure 5.1: Processing pipeline of the AI-based malaria classification system.

A complete thick blood smear image is used as an input for classification, without detecting parasite regions or identifying parasite types. The goal is to classify each image as either infected or uninfected, which is the main goal of this thesis. This differs from many of the reviewed approaches in chapter 3, which first extract parasite regions or cell-level areas before classification.

Before the image is passed to the CNN model, preprocessing is applied. This includes normalisation and resizing to the input size required by the selected architecture. During training, data augmentation is also used.

For the resolution experiments, the same source images are processed at different resolution levels in separate runs. The image is first reduced to the selected lower resolution and is then resized to the input size expected by the model. This makes it possible to compare how the same architecture behaves under different image resolutions.

The preprocessed image is then passed through one of the selected CNN architectures: ResNet50, ResNet18, or MobileNet. The model produces a binary classification result predicting whether the image is infected or uninfected.

## 5.3 Training Strategy

The selected CNN models are trained using transfer learning. To save time and computing power, pretrained ImageNet weights are used as the starting point for the models. The models are then fine-tuned using the malaria thick blood smear images. This approach makes it possible for the networks to start from already learned image features and adapt them to the task of malaria classification.

The final classification layer is adapted to two output classes: infected and uninfected. Cross-entropy loss, explained in Section 3.3, is used during the training to show how wrong the model's prediction is compared with the correct class. Adam is then used as the optimiser for updating the model parameters during training. The same training strategy is applied to ResNet50,

ResNet18, and MobileNet in order to keep the comparison between the architectures more fair.

To support model generalisation, data augmentation is applied only to the training images. The augmentation includes horizontal and vertical flipping as well as small image rotations. The goal is to increase the variation of the training data and reduce the risk of overfitting. The validation images are not augmented.

During training, the validation accuracy is calculated after each epoch. Each time a higher validation accuracy is reached, the corresponding checkpoint is saved. If the same validation accuracy is reached again, the previously saved checkpoint is kept. The checkpoint with the highest validation accuracy is later used for the final test evaluation.

## 5.4 Hyperparameter Selection

The hyperparameters were selected based on the available hardware. Since the training was performed on a MacBook Pro with an M2 chip and 8 GB of memory, moderate settings were chosen to keep the training stable and manageable.

The same main hyperparameters were used initially for ResNet50, ResNet18, and MobileNet. A learning rate of 0.0001, which is suitable for fine-tuning pretrained models, was used together with the Adam optimiser. The batch size was set to 8, and each model was trained for 15 epochs. The pretrained networks were fine-tuned as complete models rather than keeping the earlier layers frozen.

Using the same training parameters for all three architectures makes the comparison more reliable by reducing the effect of different hyperparameter settings on the results. It is also planned that, if one of the lightweight models shows very low performance, an additional optimisation may be carried out. This approach was also discussed within the project team and was considered suitable for a fair comparison of the selected architectures.

After each epoch, the validation accuracy was calculated, and the checkpoint with the highest validation accuracy was saved for later evaluation.

Table 5.1: Main hyperparameters used for training.

| **Hyperparameter** | **Selected value** |
|---|---|
| Learning rate | 0.0001 |
| Batch size | 8 |
| Number of epochs | 15 |
| Optimizer | Adam |
| Weight initialization | pretrained on ImageNet |
| Fine-tuning | Complete network |
| Model selection | Highest validation accuracy |

## 5.5 Evaluation Setup

The evaluation is constructed to compare the selected CNN architectures for this work under the same conditions and to investigate how reduced image resolution affects the classification performance.

The high-resolution baseline uses an image size of $224 \times 224$ pixels, which corresponds to the standard input size used by the pretrained ResNet and MobileNet models. For the reduced resolution experiments, the same source images are reduced to lower resolutions (LR1/LR2/LR3)

and then resized back to $224 \times 224$ pixels before being given to the CNN model. In this way, the input size of the network stays unchanged while the amount of image detail is reduced. The lowest tested resolution is $56 \times 56$ pixels and represents the extreme low-resolution case.

The dataset is divided into training, validation, and test sets using a standards patient-based split of 70%, 15%, and 15%. All images belonging to the same patient remain in the same subset. This avoids using images from the same patient for both training and evaluation.

The classification performance is evaluated using accuracy, sensitivity, specificity, precision, F1-score, and ROC-AUC. These metrics were already explained in Section 3.3. Using several metrics makes it possible to evaluate the overall classification accuracy as well as the behaviour of the models for infected and uninfected images.

The experiments are carried out in several steps. First, a high-resolution baseline is established. The influence of reduced image resolution is then investigated using lower-resolution runs of the same images. At the end, the selected CNN architectures are compared under the defined evaluation conditions. This approach reduces the number of unnecessary training runs while allowing the influence of image resolution to be investigated.

Table 5.2: Resolution levels used in the evaluation.

| **Resolution level** | **Image size** |
|---|---|
| HR | 224×224 pixels |
| LR1 | 160×160 pixels |
| LR2 | 112×112 pixels |
| LR3 | 56×56 pixels |

## 5.6 Grad-CAM Analysis

During the test phase, Grad-CAM is used to understand the behaviour of models that show weak performance. As discussed in Section 3.10, Grad-CAM generates a heatmap that highlights the image regions that have the strongest influence on a prediction.

The analysis is therefore focused on underperforming cases, for example models or resolution settings with lower classification performance. The goal here is to investigate whether the model still focuses on meaningful regions inside the microscopic image, for example areas containing parasites, or whether the prediction is influenced by unrelated areas such as the background or image edges.

Grad-CAM is also applied to some selected true-positive, true-negative, false-positive, and false-negative predictions. This makes it possible to compare successful and incorrect classifications and to better understand how the model behaves in different prediction cases.

The Grad-CAM analysis is used as a supporting method together with the evaluation metrics. This helps to identify reasons for unusual model behaviour that cannot be explained by accuracy, sensitivity, specificity, or F1-score alone. The generated heatmaps are presented and discussed in the evaluation chapter.

# 6 Implementation

In this chapter, it is described how the AI-based malaria classification system was implemented based on the concept developed in Chapter 5. It explains the dataset used for the implementation, the preprocessing pipeline, the software tools (such as PyTorch, torchvision and Matplotlib), and the structure of the developed pipeline. This chapter also explains how the individual components (dataset loading, preprocessing, training, testing and Grad-CAM Analysis) are connected and how reproducibility is supported through fixed data splits, saved model checkpoints, and version-controlled development.

## 6.1 Data Acquisition

The image data used for the implementation was obtained from publicly available malaria microscopy datasets. No new blood smear images were collected as part of this thesis.

The training, validation, and main evaluation were based on the NLM/NIH thick blood smear dataset, which has also been used by Kassim et al. [22]. It was decided that the NLM/NIH dataset was the perfect match for the developed model because it contains thick blood smear images from infected and uninfected patients, which means it fits the binary classification task of this thesis. Another important advantage is its patient-based structure, which makes it easy to separate the data at patient level and avoid images from the same patient appearing in both training and evaluation sets.

Two additional public datasets were considered during the dataset selection. The Uganda microscopy dataset described by Quinn et al. [21] contains thick blood smear images with parasite annotations and is more suitable for parasite detection and localisation tasks. This dataset is available for download on the official website of AIR Uganda which belongs to Makerere University. The Uganda dataset was later used only for an external test of the trained models to check whether the predictions depend strongly on dataset-specific image characteristics. The second dataset is the Vall d'Hebron dataset [23] which also contains labelled thick blood smear images, but mainly provides malaria-positive samples. This one is also not suitable for the infected-versus-uninfected classification used in this work. To get access to this dataset, an application was submitted, and access to the dataset was granted afterwards.

The Vall d'Hebron dataset was obtained as a possible data source for a future extension of the project, for example towards parasite detection or species identification.

No image data from the Ethiopian Artificial Intelligence Institute were available for the implementation of this thesis. The developed AI method and the other components of the overall project are intended to be transferred to EAII for later development and testing.

## 6.2 Dataset Description

As explained in the previous section, the NLM dataset is the most suitable dataset for this work. It consists of three thick blood smear subsets which contain images from uninfected patients, patients infected with Plasmodium vivax, and patients infected with Plasmodium falciparum. All these three subsets contain Giemsa-stained thick blood smear images acquired through the eyepiece of a microscope at 100× magnification using a smartphone camera. The original images are RGB images with a resolution of 3024 × 4032 pixels [8, 22].

The uninfected subset contains 1141 images from 50 malaria-negative patients [22]. The images are stored in TIFF format and are organised in separate folders by patient. The dataset also provides annotations for white blood cells, which are not needed for the whole-image classification used in this thesis.

The P. vivax subset contains images from 150 malaria-infected patients, with approximately 20 images per patient [22]. Kassim et al. report in their paper a total of 3013 images, while 3014 images were found in the downloaded dataset and used in the implementation. The images are stored in JPEG format and parasite locations are provided in separate annotation files. Since the goal of this thesis is to classify complete images rather than identify individual parasites, these parasite annotations were also not used for the classification process.

The P. falciparum subset contains images from 150 infected patients, with approximately 12 images per patient [8]. Yang et al. report a total of 1819 images, while 1883 images were found in the downloaded dataset and used in the implementation. These images are also stored in JPEG format. The corresponding annotation files contain information about parasite and white blood cell positions. Just like the P. vivax subset, these annotations were not required for the whole-image classification task.

For the classification task, the two infected subsets were combined into one positive class. Images from patients infected with P. vivax or P. falciparum were assigned to the class infected, while the images from the uninfected subset were assigned to the class uninfected. In total, 6038 images were included in the implementation: 1141 uninfected images, 3014 P. vivax images, and 1883 P. falciparum images. The final classification task therefore contains two output classes: infected and uninfected.

In addition to the NLM dataset, the Uganda microscopy dataset was used as an external test dataset [21]. It contains 1182 malaria-positive images stored in JPEG format. The dataset does not provide patient IDs, and all images used for this external evaluation were assigned to the test split. The Uganda dataset was kept completely separate from the training and validation process and is therefore not included in the 6038 NLM images described above.

## 6.3 Data Preprocessing Pipeline

Before diving into training the model, the raw data must first be prepared in a way that makes it readable and consistent. This section walks through the preprocessing steps used in this work.

The preprocessing pipeline was implemented in the following steps: First, the image information and binary labels were prepared using the script **build_labels.py**. The images were then loaded using the **MalariaDataset** class from the **dataset.py** script. After that, the required preprocessing transforms were applied for training, validation, and testing. These transforms are different depending on the split. Training images received random augmentation, while validation and test images did not. For the low-resolution experiments, the same images were reduced to the selected resolution before being resized back to $224 \times 224$ pixels, which is the required input resolution for the model.

The script **build_labels.py** goes through all three NLM image folders for **Uninfected**, **P. vivax**, and **P. falciparum** images. The binary labels are assigned based on the images folder. Images from the uninfected subset receive label 0, while images from both infected subsets receive the same label 1. As mentioned in Section 6.2, the parasite and white blood cell annotation files provided with the dataset are not used, since the classification in this thesis is based on the complete image.

A file named **labels.csv** is generated by this script, which stores the following information for every image: the image path, the class, the binary label, the patient ID, and the assigned dataset split. The split used here is a conventional split using 70 % of the patients for training, 15 % for validation, and 15 % for testing with a random seed of 42. The output is 4186 training images, 899 validation images, and 953 test images.

This generated labels file is then read by the **MalariaDataset** class in **dataset.py**. This class selects the images based on the chosen split (train, validation or test) and also adds the stored path from the CSV file to the main dataset path. Each image is then opened by PIL (Pillow) and converted to RGB format. The binary label is read directly from the generated CSV file. This class does not define the resizing, augmentation, or normalisation steps. The transform chain is passed to the class from the training or testing script.

For the high-resolution experiment, no additional resolution reduction is needed. This means, the original images are resized directly to $224 \times 224$ pixels before being given to the CNN model. During training, random horizontal and vertical flips are used with a probability of 0.5. In addition, random rotations between $-15°$ and $+15°$ are also applied. After the augmentation, the image is converted to a numerical format (PyTorch tensor) and normalised using the ImageNet mean and standard deviation values. These preprocessing steps are defined directly in the **train.py** script.

The same preprocessing pipeline is performed on the low-resolution experiments, with an additional resizing process before the augmentation. The image is first resized to the resolution defined for the experiment and is then resized back to $224 \times 224$ pixels. LR1 uses $160 \times 160$ pixels, LR2 uses $112 \times 112$ pixels, and LR3 uses $56 \times 56$ pixels. The final input size of the CNN remains $224 \times 224$ pixels for all resolutions.

The tested resolution levels are summarised in Table 6.1.

Table 6.1: Resolution levels used in the experiments.

| **Resolution level** | **Reduced resolution** | **CNN input size** |
|---|---|---|
| HR | No reduction | $224 \times 224$ |
| LR1 | $160 \times 160$ | $224 \times 224$ |
| LR2 | $112 \times 112$ | $224 \times 224$ |
| LR3 | $56 \times 56$ | $224 \times 224$ |

To avoid confusion, these low-resolution images are not images that were originally captured at these resolutions. Instead, the lower-resolution conditions are simulated from the same NLM dataset images. The images are first reduced and then resized back to $224 \times 224$ pixels. In short, the network input size remains unchanged while image details are reduced.

The Figure 6.1 shows an example of the difference between the HR and LR3 conditions.

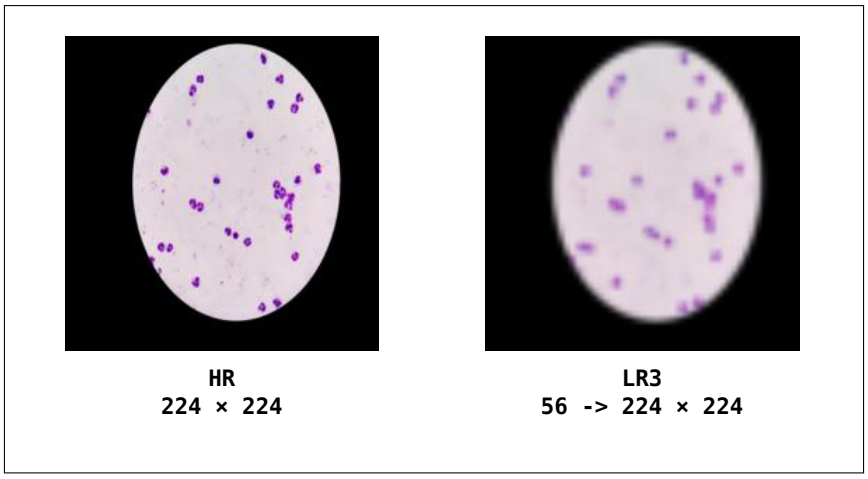



Figure 6.1: Example of the same thick blood smear image from patient folder **PvTk1** under the HR and LR3 resolution conditions.

Keeping this preprocessing step simple was also important. As a result, the resolution reduction was implemented directly in the preprocessing pipeline.

Listing 6.1: Resolution reduction used for the LR3 experiment.

```
1 transforms.Resize((56, 56)),
2 transforms.Resize((224, 224)),
```

The same implementation is used for LR1 and LR2 by replacing the reduced resolution with $160 \times 160$ pixels and $112 \times 112$ pixels.

For validation and testing, no random augmentation is used. In the HR experiment, the images are only resized to $224 \times 224$ pixels, converted to tensors, and normalised. In the case of a low-resolution experiment, the same resolution reduction used during training is also applied to the validation and test images before resizing them back to $224 \times 224$ pixels. After that, the images are converted to tensors and normalised. This means that the same resolution condition is used for training, validation, and testing, while the random flips and rotations are only used during training to keep the results reliable.

For the external test with the Uganda dataset, the images were processed using the same LR3 test transform, first resized to $56 \times 56$ pixels and then resized back to $224 \times 224$ pixels. The already trained models were used directly without any retraining.

## 6.4 Software Stack and Tools

This work was developed in Python 3.12.2 and executed through the macOS Terminal. PyTorch was used as the main deep learning framework to build, train, and test the CNN models. Additionally, the torchvision library provided the pretrained network architectures (ResNet50, ResNet18 and MobileNetV3-Small) and also the image preprocessing transforms.

The microscope images were loaded by Pillow(PIL), while pandas read and handled the **labels.csv** file. The NumPy library managed the numerical operations, and to evaluate the developed models, scikit-learn was used to calculate metrics such as the confusion matrix, classification report, and ROC-AUC score. Matplotlib created the training curves, confusion matrices, and ROC curves.

When it comes to understanding how some models behave, the **pytorch-grad-cam** library plays a big role. It was used to generate Grad-CAM and Grad-CAM++ visualisations, which show which regions the model is concentrating on while making a prediction.

The scripts were also prepared to run directly from the Terminal. The Python **argparse** module is responsible for passing settings such as the dataset path, number of epochs, batch size, learning rate, and checkpoint folder as command-line arguments. The `-data-root` argument replaced the dataset path that was previously written directly in the script. This made the script more flexible, since it can be used with different dataset locations or on another computer without changing the path inside the Python file.

Listing 6.2: Command-line arguments used in the training script.

```
def get_args():
    parser = argparse.ArgumentParser()
    parser.add_argument("--data-root", type=str, required
        =True)
    parser.add_argument("--labels-csv", type=str, default
        ="labels.csv")
    parser.add_argument("--epochs", type=int, default=15)
    parser.add_argument("--batch-size", type=int, default
        =8)
    parser.add_argument("--lr", type=float, default
        =0.0001)
    parser.add_argument("--checkpoint-dir", type=str,
                        default="checkpoints")
    return parser.parse_args()
```

**Type and extent of AI use**
`argparse` was first seen on some Kaggle competitions inside a python script. The idea sounded useful and Claude was then asked to create a similar function based on the variables used in this work. Using the –data-root argument was also suggested by Claude instead of keeping the dataset path fixed inside the script. Both were added to this project.

**The individual contribution**
The function was then integrated into the training script with the parameters that were used for the experiments, such as dataset path, epochs, batch size, learning rate, and checkpoint folder. The fixed dataset path was replaced with the `-data-root` argument.

**Critical evaluation of the AI contribution**
The `argparse` function is not necessary for the model, because the values could also be written directly in the script. But it made the work more flexible because the settings can be changed from the Terminal without going into the Python file every time. The `-data-root` argument was especially useful because the same script can now be used with a different dataset location without changing the source code.

The training and testing scripts also include an automatic device selection. The scripts first check if a CUDA-supported GPU is available. If not, they check for the Apple MPS backend, and if neither is available, the CPU is used. Since the experiments in this work were executed on the MacBook Pro M2, the MPS backend was selected. The automatic device-selection logic was adapted from an example found on the PyTorch Forums.[1]

[1] https://discuss.pytorch.org/t/same-model-gets-slightly-different-results-on-different-platforms/194551, accessed 24 July 2026.

Listing 6.3: Automatic device selection adapted from a PyTorch Forum example.

```
if torch.cuda.is_available():
    device = torch.device("cuda")
elif torch.backends.mps.is_available():
    device = torch.device("mps")
else:
    device = torch.device("cpu")

print("using device:", device)
```

A GitHub repository was also created to make access to the model architectures and scripts easy for all team members, serving as a platform for version control, source code storage, and project updates.

The software versions used during the implementation are shown in Table 6.2.

Table 6.2: The software and library versions used in the implementation.

| **Software / Library** | **Version** |
|---|---|
| Python | 3.12.2 |
| PyTorch | 2.13.0 |
| torchvision | 0.28.0 |
| pandas | 2.2.2 |
| NumPy | 1.26.4 |
| Pillow | 10.4.0 |
| scikit-learn | 1.5.1 |
| Matplotlib | 3.10.0 |
| pytorch-grad-cam | 1.5.5 |

## 6.5 Pipeline Architecture and Integration

The implementation of the classification model is made of separate scripts, where each script is responsible for a specific stage of the overall pipeline. The pipeline consists of **build_labels.py**, **dataset.py**, **train.py**, **test.py**, and **gradcam.py**. Keeping these scripts separated made it easier to reuse the same parts of the pipeline for different network architectures and also for reduced resolutions.

The pipeline starts with the NLM dataset folders, which offer a clean separation of the two classes (one folder for uninfected and 2 folders for infected: P. vivax and P. falciparum). The first script is **build_labels.py** that goes through the three image folders and creates the **labels.csv** file. This CSV file stores the image path, class, binary label, patient ID, and the assigned dataset split. These generated information are then used by the **MalariaDataset** class in the second script which is **dataset.py**. This **MalariaDataset** class links the information from the CSV file to the actual image files and then loads the requested split (training, validation, or test).

The **dataset.py** script does not contain the preprocessing for the specific experiments. Instead, the preprocessing transforms are provided to the **MalariaDataset** class from the training or testing script. This means that the same dataset class can be reused for the HR, LR1, LR2, and LR3 experiments by only changing the corresponding transform pipeline.

During training, **train.py** loads the training and validation data through the defined **MalariaDataset** class. After that, the chosen CNN is configured for binary classification and trained with the given hyperparameters. After every epoch, the model is also evaluated on the validation set.

After each validation step, the model is saved as **best_model.pth** if its accuracy outperforms all previous epochs. This model represents the best model found during the training process.

After training, **test.py** loads the same CNN architecture used in training and the saved **best_model.pth** checkpoint. The fixed test split is used for the final evaluation. During testing, the model parameters are not changed. The script calculates the evaluation metrics (accuracy, sensitivity, specificity, precision, f1-score and ROC-AUC) and also creates outputs such as the confusion matrix and ROC curve.

The main connection between the scripts and the generated files is shown in Figure 6.2. The external test was not included in the figure because it uses a separate dataset and a separate testing script and does not form part of the main NLM training, validation, and testing pipeline.

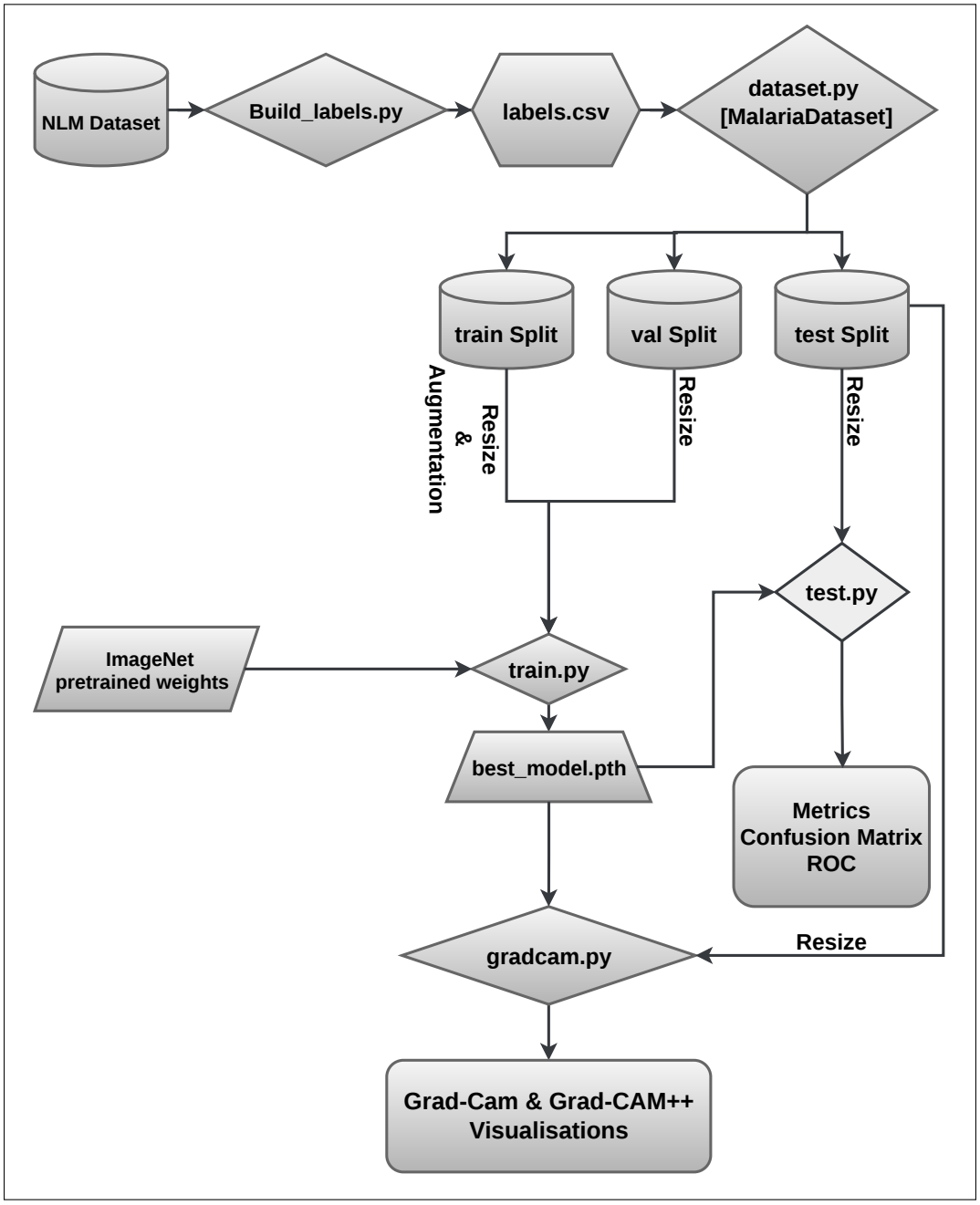



Figure 6.2: Script and data flow of the implemented classification pipeline.

For the external test, the same testing process was reused with a separate **labels_uganda.csv** file and the script **test_uganda.py**. The Uganda images were kept separate from the main NLM pipeline and were only used with already trained checkpoints, without any additional training or optimisation.

The last analysis step is conducted by **gradcam.py**. This script loads the already trained checkpoint from the training and uses images selected from the test set to generate Grad-CAM and Grad-CAM++ visualisations. In the **gradcam.py** script, the test predictions are first separated into true positive, true negative, false positive, and false negative groups. A limited number of examples from these groups are then used for the explainability analysis. The generated heatmaps are saved as image files for later visual evaluation and also for documentation purposes.

To keep the completed experiments separated from each other, each experiment was stored in its own folder together with the corresponding scripts and also the generated outputs. This made it easier to return to an older experiment without changing its original settings or results. It also helped when comparing the different architectures and resolution conditions. Furthermore, this separation allowed the team members to access the finished experiments without any complication.

The number of experiments was not the same for all three network architectures. ResNet50 was trained and tested at all four resolution conditions: HR, LR1, LR2, and LR3. ResNet18 and MobileNetV3-Small were trained and tested only at HR and LR3. This resulted in a total of eight main experiment runs. The different experiment runs are shown in Table 6.3.

Table 6.3: Experiment runs for the three CNN architectures.

| **Architecture** | **HR** | **LR1** | **LR2** | **LR3** |
|---|---|---|---|---|
| ResNet50 | Yes | Yes | Yes | Yes |
| ResNet18 | Yes | No | No | Yes |
| MobileNetV3-Small | Yes | No | No | Yes |

After completing the four resolution experiments with ResNet50, only two resolutions (HR and LR3) were selected for the comparison with ResNet18 and MobileNetV3-Small. These two conditions represent the two ends of the tested resolution range. HR represents the normal input condition, while LR3 with $56 \times 56$ pixels represents the highest simulated resolution reduction used in this work. This made it possible to compare the three architectures under normal and strongly reduced image quality without repeating all four resolution experiments for every architecture.

The LR3 condition is relevant to the planned use of the model together with the smartphone-based ocular adapter. However, the $56 \times 56$ pixel resolution is not a measured resolution of the final adapter. It was used only as a simulated worst-case resolution to investigate the behaviour of the models when the image detail is extremely reduced.

Even though performance degraded across all architectures at LR3, the drop in specificity was the most obvious. Because MobileNet is lightweight and more suitable for future deployment, it was decided to optimise only this model. The original performance metrics for ResNet18 and ResNet50 were preserved as baseline comparisons.

The LR3 experiment on MobileNetV3-Small was then performed in three stages. The first stage used the same settings as the other architectures (ResNet50 and ResNet18): a learning rate of 0.0001, a batch size of 8, and 15 epochs. The training behaviour of this configuration was unstable.

For the first optimisation, the learning rate was reduced to 0.00003, the batch size was increased to 16, and the number of epochs was also increased to 30. These changes resulted in more stable

training. After this optimisation, the classification was still not balanced enough between the two classes. That is why a second optimisation was performed by keeping the previous settings from the first optimisation and additionally adding class weights of $[2.0, 1.0]$ for the uninfected and infected classes. This made errors on the uninfected class have a higher influence on the training loss.

The three MobileNetV3-Small configurations are summarised in Table 6.4.

Table 6.4: MobileNetV3-Small configurations used for the LR3 experiment.

| Configuration | Learning rate | Batch size | Epochs | Class weights |
|---|---|---|---|---|
| Baseline | 0.0001 | 8 | 15 | None |
| Optimisation 1 | 0.00003 | 16 | 30 | None |
| Optimisation 2 | 0.00003 | 16 | 30 | [2.0, 1.0] |

This optimisation was deliberately limited to MobileNetV3-Small. As already discussed in Chapter 5, the ResNet models are mainly used as stronger reference architectures for comparing classification performance, while MobileNet is more relevant for lightweight deployment. Re-optimising all three architectures was therefore outside the scope of this work. The numerical results of all experiments and the results of the MobileNet optimisations are presented later in Chapter 7.

## 6.6 Reproducibility

The source code of the project is maintained in a GitHub repository and is available at `https://github.com/H-Yassine/malaria-diagnosis-master-thesis.git`.

The final version of the implementation used in this thesis corresponds to commit **`https://github.com/H-Yassine/malaria-diagnosis-master-thesis/commit/b256db2`**. The repository history contains **24 commits**, which document the development of the implementation between july 2026 and october 2026. In addition to the source code, the repository contains a **meetings/** folder where presentation files and project updates from the regular meetings were kept. The image datasets themselves are not distributed through the repository. The repository therefore provides only the code and pipeline structure, while the original datasets have to be obtained separately from their corresponding sources.

The structure of the repository is shown in Figure 6.3. The project files are separated into four main folders: data, meeting documents, thesis files, and implementation scripts. Every experiment folder is stored inside the scripts folder of the repository.

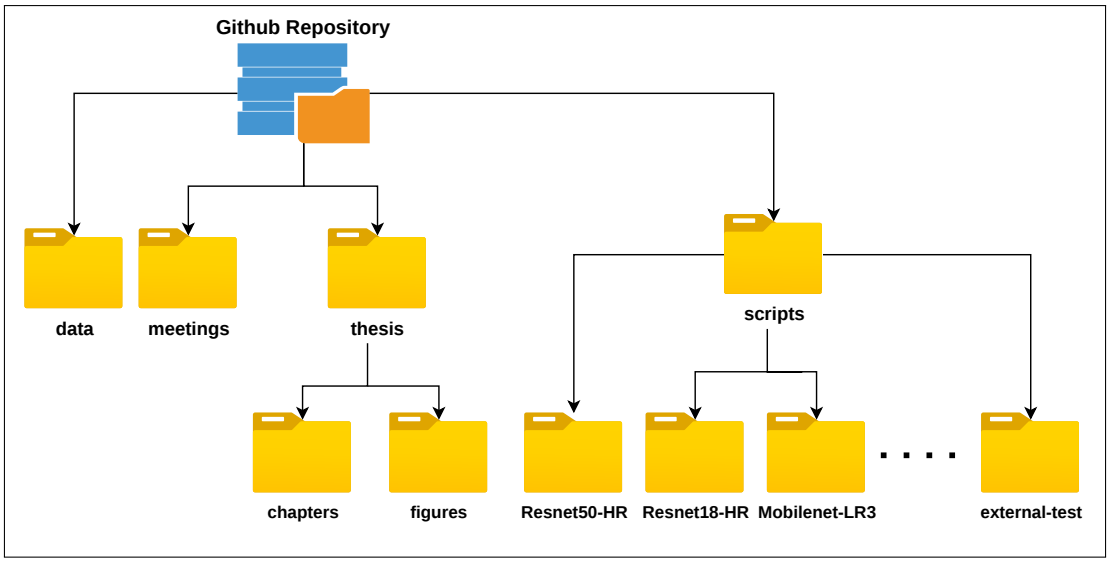



Figure 6.3: Overview of the GitHub repository structure.

The different experiments were organised in separate folders. Each of the eight main experiments contains the corresponding **dataset.py**, **train.py**, **test.py**, **labels.csv**, and the generated experiment outputs. Once an experiment was completed, its folder was kept unchanged so that later modifications did not affect already finished experiments. The Grad-CAM and Grad-CAM++ scripts were stored inside the corresponding experiment folders, while the external Uganda test was kept separately in its own **external-test/** folder.

Not all paths were handled in exactly the same way. For example, in **gradcam.py**, the dataset path and checkpoint path are defined at the beginning of the script, while other scripts use command-line arguments for some paths. This reflects the way the individual experiment scripts were developed and stored during the project.

An important part of the reproducibility is the fixed dataset split. The **build_labels.py** script performs the 70/15/15 split at patient level and uses a fixed random seed of 42. This means that running the script again with the same dataset produces the same patient assignment for training, validation, and testing. The resulting assignments are stored in **labels.csv** and reused throughout the experiments. Each experiment also keeps its **best_model.pth** checkpoint together with its result files and visual outputs. Information about the datasets is additionally documented in a **README.md** file inside the **data/** folder.

There are still some limits to reproducibility. The fixed seed of 42 controls the patient split, but the training scripts do not fix all random operations with a global PyTorch seed. Random data augmentation and model training can therefore produce small differences when an experiment is trained again. The experiments were also executed using the Apple MPS backend, where exact numerical results are not guaranteed to be identical between repeated runs. For this reason, the saved checkpoints and result files are kept together with each completed experiment. In addition, the external image datasets cannot be redistributed with the source code and have to be downloaded separately from their original providers.

# 7 Testing and Evaluation

This chapter presents the results of the developed models and evaluates how they perform at the different resolution levels. In addition to the normal test results, Grad-CAM, Grad-CAM++, and an external test dataset were also used to better understand the models behaviour.

## 7.1 Experimental Setup

To keep the evaluation reliable, the same patient-based dataset split described in Section 6.2 was used for all experiments. The NLM dataset was divided into training, validation, and test sets using a 70/15/15 ratio with a fixed random seed of 42. This resulted in 4186 training images, 899 validation images, and 953 test images. The split was based on patients, so images from the same patient appear in only one split.

The test set contains 750 infected images and 203 uninfected images. The number of infected samples is therefore considerably higher than the number of uninfected samples. This class imbalance is important when interpreting the results, especially sensitivity and specificity, since a high overall accuracy alone does not show how well both classes are recognised.

A total of eight main experiments were performed. ResNet50 was evaluated at all four resolution conditions (HR, LR1, LR2, and LR3), while ResNet18 and MobileNetV3-Small were evaluated only at HR and LR3. After the ResNet50 experiments showed that LR1 remained very close to the HR condition, the intermediate resolution levels were not repeated for the other architectures. Instead, HR and LR3 were selected to represent the two ends of the investigated resolution range. The exact image resolutions and preprocessing steps for HR, LR1, LR2, and LR3 are shown and explained in Section 6.3.

The training settings used for the eight main experiments are summarised in Table 7.1. The same configuration was used for all eight baseline experiments, with 15 epochs, a batch size of 8, a learning rate of 0.0001, and CrossEntropyLoss. MobileNetV3-Small at LR3 was additionally optimised, as described in Section 6.5. The final configuration used 30 epochs, a batch size of 16, a learning rate of 0.00003, and a weighted CrossEntropyLoss with class weights of $[2.0, 1.0]$.

Table 7.1: Training configuration of the eight main experiments.

| **Architecture** | **Resolution** | **Epochs** | **Batch** | **LR** | **Loss** |
|---|---|---|---|---|---|
| ResNet50 | HR | 15 | 8 | 0.0001 | CrossEntropyLoss |
| ResNet50 | LR1 | 15 | 8 | 0.0001 | CrossEntropyLoss |
| ResNet50 | LR2 | 15 | 8 | 0.0001 | CrossEntropyLoss |
| ResNet50 | LR3 | 15 | 8 | 0.0001 | CrossEntropyLoss |
| ResNet18 | HR | 15 | 8 | 0.0001 | CrossEntropyLoss |
| ResNet18 | LR3 | 15 | 8 | 0.0001 | CrossEntropyLoss |
| MobileNetV3-Small | HR | 15 | 8 | 0.0001 | CrossEntropyLoss |
| MobileNetV3-Small | LR3 | 15 | 8 | 0.0001 | CrossEntropyLoss |

For the final evaluation, the saved **best_model.pth** checkpoint from each experiment was loaded and evaluated on the test split. The checkpoint was selected during training based on the validation accuracy, as described in Chapter 5. During testing, the model was set to evaluation mode and its parameters were not changed.

The reported evaluation metrics are accuracy, sensitivity, specificity, precision, F1-score, and ROC-AUC. The definitions and meaning of these metrics were already introduced in Section 3.3.

Because the model is intended for screening, sensitivity is especially important. A false negative means that an infected image is classified as uninfected, which should be avoided in a screening system. Specificity is considered together with sensitivity to show whether good detection of infected images is achieved without incorrectly classifying too many uninfected images as infected.

## 7.2 Full-Resolution Results

First, the models were compared using full-resolution (HR) images. All three architectures were evaluated on the same test set of 953 images. These results will be used as a reference before analysing the effect of reduced image resolution in the following sections.

The results are shown in Table 7.2.

Table 7.2: Test results of the three architectures at full resolution.

| Architecture | Accuracy | Sensitivity | Specificity | Precision | F1 | ROC-AUC |
|---|---|---|---|---|---|---|
| ResNet50 | 0.9958 | 0.9973 | 0.9901 | 0.9973 | 0.9973 | 0.9997 |
| ResNet18 | 0.9969 | 1.0000 | 0.9852 | 0.9960 | 0.9980 | 1.0000 |
| MobileNetV3-S | 0.9969 | 1.0000 | 0.9852 | 0.9960 | 0.9980 | 0.9999 |

All three models achieved very high results at full resolution, and the differences between them were small. ResNet50 reached an accuracy of 0.9958 and misclassified four of the 953 test images. Two infected images were classified as uninfected and two uninfected images were classified as infected.

ResNet18 and MobileNetV3-Small both reached an accuracy of 0.9969. Both models correctly classified all 750 infected images and therefore achieved a sensitivity of 1.0000. Each model incorrectly classified three of the 203 uninfected images as infected, resulting in a specificity of 0.9852.

The absence of false negatives in ResNet18 and MobileNetV3-Small is important for screening, as zero infected images were missed. In general, the differences between the three architectures are very small and are based on a few images. The HR results show that all three architectures perform similarly well under the normal resolution condition.

## 7.3 Performance under Reduced Resolution

### 7.3.1 ResNet50 across the Resolution Levels

ResNet50 was the only architecture tested at all four resolution conditions. This makes it possible to observe how the model performance changes when the image resolution is reduced from HR to LR3. The results are shown in Table 7.3.

Table 7.3: ResNet50 test results at the four resolution conditions.

| Resolution | Accuracy | Sensitivity | Specificity | Precision | F1 | ROC-AUC |
|---|---|---|---|---|---|---|
| HR | 0.9958 | 0.9973 | 0.9901 | 0.9973 | 0.9973 | 0.9997 |
| LR1 | 0.9958 | 0.9987 | 0.9852 | 0.9960 | 0.9973 | 0.9999 |
| LR2 | 0.9843 | 0.9960 | 0.9409 | 0.9842 | 0.9901 | 0.9994 |
| LR3 | 0.9286 | 0.9800 | 0.7389 | 0.9327 | 0.9558 | 0.9698 |

Figure 7.1 shows the confusion matrices for ResNet50 across the four resolution conditions.

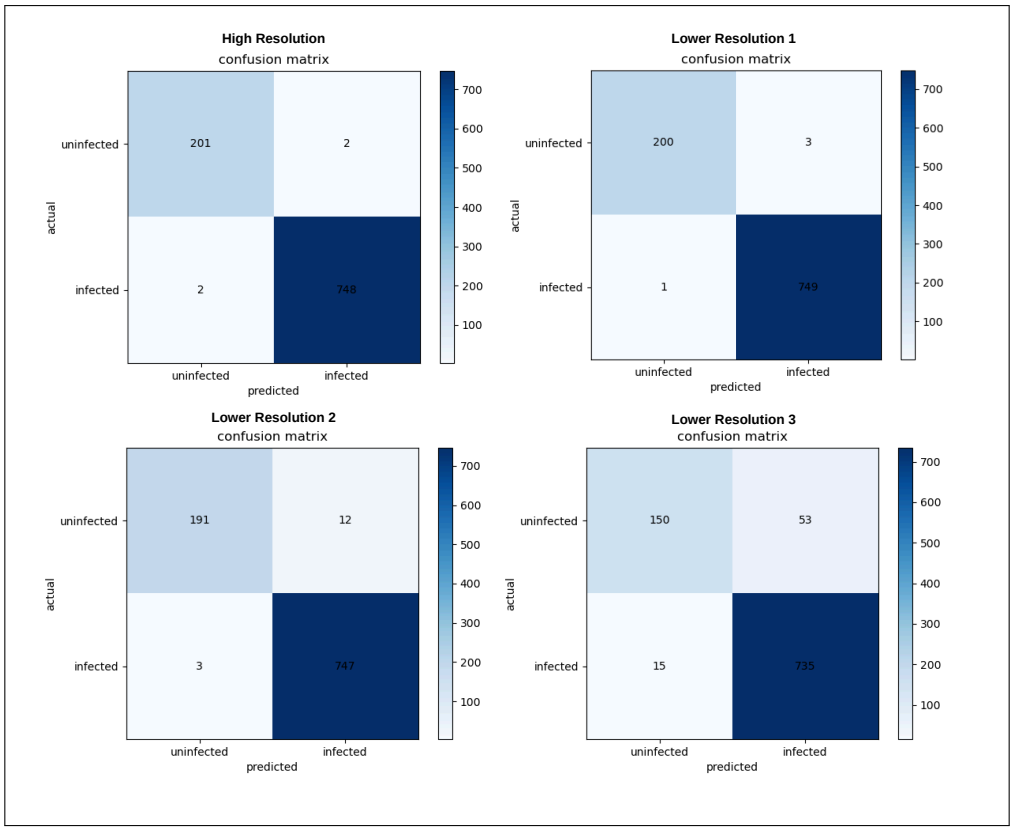



Figure 7.1: Confusion matrices of ResNet50 for HR, LR1, LR2, and LR3.

As can be seen from the results table and the confusion matrices, the overall performance stayed good under the first reduced resolution, from HR to LR1. Both resolutions reached an accuracy of 0.9958. The number of misclassified images was also the same. HR gave two false positives and two false negatives, while LR1 gave three false positives and one false negative.

At LR2, the accuracy decreased to 0.9843, while sensitivity remained high at 0.9960. The clearer change was seen in specificity, which decreased from 0.9901 at HR to 0.9409 at LR2. In LR2, 12 uninfected images were classified as infected and three infected images were classified as uninfected.

LR3 showed the strongest reduction in performance. Accuracy decreased to 0.9286 and specificity dropped to 0.7389. The model incorrectly classified 53 uninfected images as infected and missed 15 infected images. Sensitivity still remained relatively high at 0.9800, but the number of false positives increased clearly compared with previous resolutions.

In general, ResNet50 remained very stable between HR and LR1. The performance started to decrease more clearly at LR2 and especially at LR3. The largest change across the resolution levels was seen in specificity, while sensitivity stayed high across all four conditions.

### 7.3.2 Architecture Comparison at LR3

To analyse how strong resolution reduction impacts model performance, all three architectures were compared at LR3. For MobileNetV3-Small, the baseline result before optimisation was used in this comparison. The results are shown in Table 7.4.

Table 7.4: Comparison of the three architectures at LR3.

| Architecture | Accuracy | Sensitivity | Specificity | Precision | F1 | ROC-AUC |
|---|---|---|---|---|---|---|
| ResNet50 | 0.9286 | 0.9800 | 0.7389 | 0.9327 | 0.9558 | 0.9698 |
| ResNet18 | 0.9108 | 0.9453 | 0.7833 | 0.9416 | 0.9434 | 0.9666 |
| MobileNetV3-S | 0.8930 | 0.9480 | 0.6897 | 0.9186 | 0.9331 | 0.9378 |

At LR3, the differences between the three architectures are clearer. ResNet50 achieved the highest accuracy of 0.9286 and the highest sensitivity of 0.9800. It missed 15 infected images and incorrectly classified 53 uninfected images as infected.

ResNet18 reached an accuracy of 0.9108. The sensitivity was lower at 0.9453, with 41 infected images classified as uninfected. On the other hand, it achieved the highest specificity of the three models at 0.7833 and gave 44 false positives.

The baseline MobileNetV3-Small reached an accuracy of 0.8930 and a sensitivity of 0.9480. It produced 39 false negatives and 63 false positives. Its specificity of 0.6897 was the lowest of all the three architectures.

In summary, all three architectures showed a reduction in performance at LR3 compared with the full-resolution results. ResNet50 kept the highest accuracy and sensitivity, while ResNet18 achieved the highest specificity. MobileNetV3-Small showed the lowest baseline performance at LR3.

Since MobileNetV3-Small was the architecture considered for lightweight deployment, a two-stage optimisation was investigated in the following section.

## 7.4 MobileNet Optimisation at LR3

Considering that MobileNetV3-Small was selected as a lightweight architecture for future deployment, its performance at LR3 was investigated further. The baseline model showed lower performance than the two ResNet architectures, especially in specificity. Therefore, two optimisation stages were performed.

In the first optimisation stage, the learning rate was reduced from 0.0001 to 0.00003, the batch size was increased from 8 to 16, and the number of training epochs was increased from 15 to 30. In the second stage, the same settings were kept and class weights of $[2.0, 1.0]$ were added to the CrossEntropyLoss.

The results of the three stages are shown in Table 7.5.

Table 7.5: MobileNetV3-Small results at LR3 before and after optimisation.

| Stage | Accuracy | Sensitivity | Specificity | Precision | F1 | ROC-AUC |
|---|---|---|---|---|---|---|
| Baseline | 0.8930 | 0.9480 | 0.6897 | 0.9186 | 0.9331 | 0.9378 |
| Optimisation 1 | 0.9234 | 0.9920 | 0.6700 | 0.9174 | 0.9532 | 0.9711 |
| Optimisation 2 | 0.9276 | 0.9640 | 0.7931 | 0.9451 | 0.9545 | 0.9506 |

The first optimisation stage increased the accuracy from 0.8930 to 0.9234. The sensitivity also increased from 0.9480 to 0.9920, reducing the number of false negatives from 39 to only 6. The specificity decreased slightly from 0.6897 to 0.6700, and the number of false positives increased from 63 to 67.

Because of the class imbalance in the dataset, class weights were added in the second optimisation stage. This improved the accuracy to 0.9276. The biggest improvement was seen in specificity, which increased to 0.7931 as false positives dropped to 42. Meanwhile, sensitivity reached 0.9640.

The second optimisation stage gave a more balanced result between sensitivity and specificity. It also achieved the highest accuracy, precision, and F1-score of the three MobileNetV3-Small LR3 experiments. Although the sensitivity was lower than in the first optimisation stage, it was still higher than the baseline result.

## 7.5 Explainability: Grad-CAM and Grad-CAM++

To have a better understanding of the models' performance and which image regions influenced the model predictions, Grad-CAM and Grad-CAM++ were used for some selected test images. The analysis included examples from all four groups which are the true-positive, true-negative, false-positive, and false-negative.

### 7.5.1 ResNet50 at HR and LR3

The first visual comparison was to compare the behaviour of ResNet50 at full-resolution and LR3. Figure 7.2 shows the Grad-CAM results at full resolution.

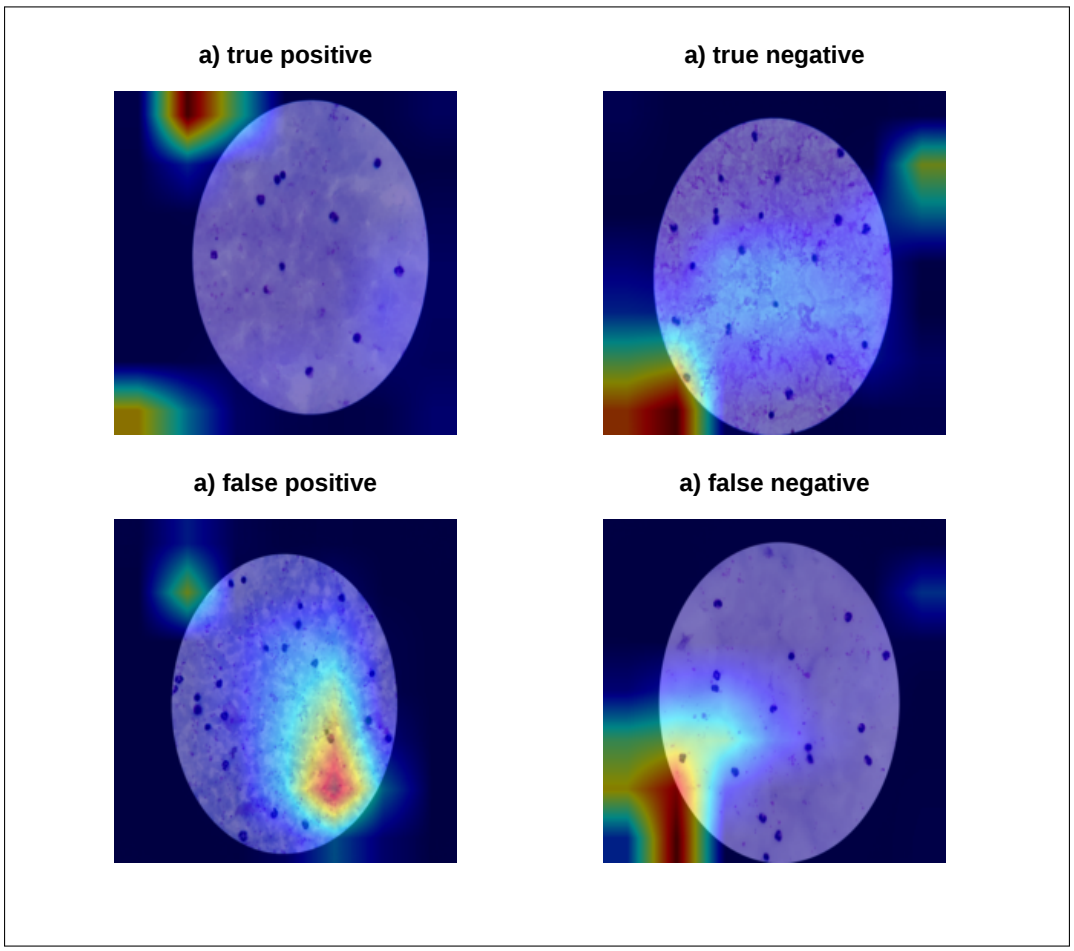



Figure 7.2: Grad-CAM results for ResNet50 at full resolution.

As observed in the Grad-CAM visualisations, the activation was not always inside the microscope field. In several examples, stronger activation was visible around the circular image boundary and in the surrounding dark background. False-positive cases showed especially stronger activation inside the blood-smear region.

To investigate these results further, the same analysis was then performed for ResNet50 at LR3 using both Grad-CAM and Grad-CAM++, to check whether similar activation patterns could be observed with both methods. Figure 7.3 compares Grad-CAM and Grad-CAM++ for the same LR3 predictions.

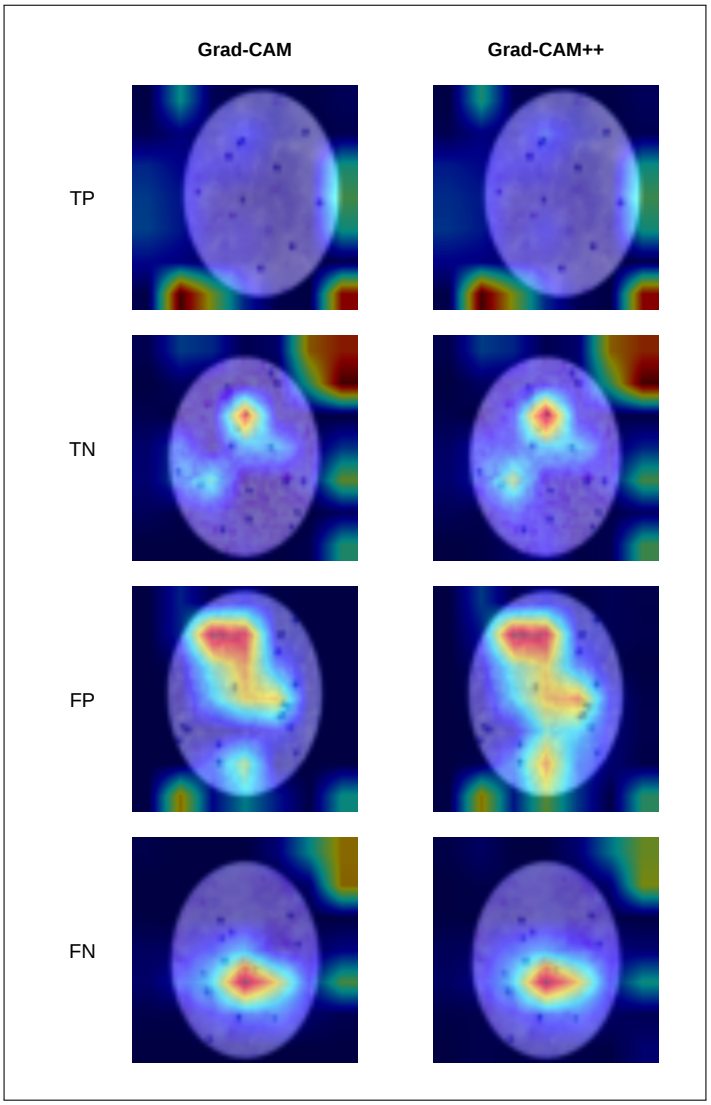



Figure 7.3: Comparison of Grad-CAM and Grad-CAM++ for ResNet50 LR3 predictions.

Looking at LR3 visualisations, activation around the circular boundary and the surrounding background was again visible in several examples. This was more noticeable in true-positive and true-negative cases. On the contrary, some false-positive and false-negative examples showed stronger activation inside the microscope field.

Grad-CAM++ showed a similar behaviour pattern. In some images, the activated regions were stronger, but activation around the circular boundary was still there. These results suggest that ResNet50 may have used image characteristics outside the blood-smear region.

### 7.5.2 Architecture Comparison at LR3

To determine whether the results observed in ResNet50 occur across all models, the same analysis was also performed for ResNet18 and MobileNetV3-Small at LR3 to check whether similar activation patterns could be observed in the other architectures.

As shown in Figure 7.4, both Grad-CAM and Grad-CAM++ showed broad activation regions in several examples for ResNet18. What is more clear is the strong activation around the circular microscope boundary and the dark surrounding area, especially in true-positive and true-negative cases. Some of the false-positive and false-negative examples showed stronger activation inside the microscope field. Similar to ResNet50, Grad-CAM++ sometimes produced more focused regions, but the general pattern stays the same.

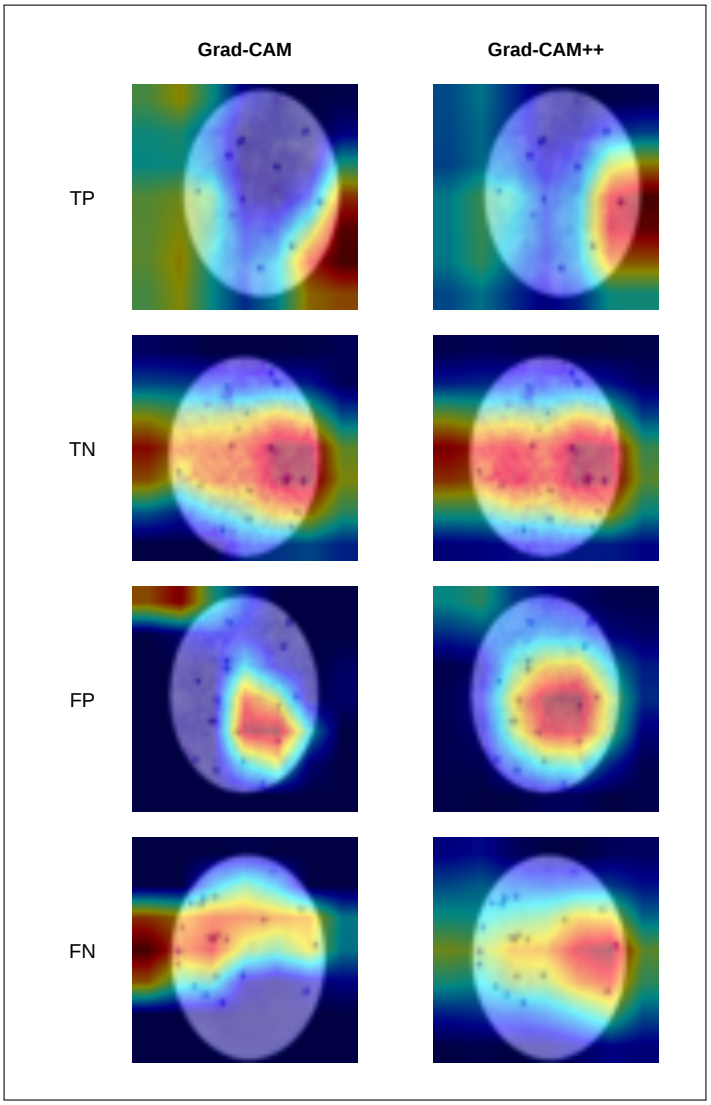



Figure 7.4: Comparison of Grad-CAM and Grad-CAM++ for ResNet18 LR3 predictions.

The activation pattern seen in ResNet50 and ResNet18 was also seen in the optimised MobileNetV3-Small. The Grad-CAM results in Figure 7.5 showed activation around the circular image boundary. The false-negative examples were mostly focused on the edges, while the false-positive examples showed stronger activation inside the microscope field. The true-negative examples were also focused outside the circular field or around its boundary. The true-positive examples showed a combination of activation inside the microscope field and also around the edges.

As expected, the Grad-CAM++ results showed similar results. The false-positive examples remained mainly focused inside the microscope field, and most true-positive examples also showed stronger activation inside the circular region. However, several false-negative and true-negative examples still contained strong activation around or outside the circular boundary.

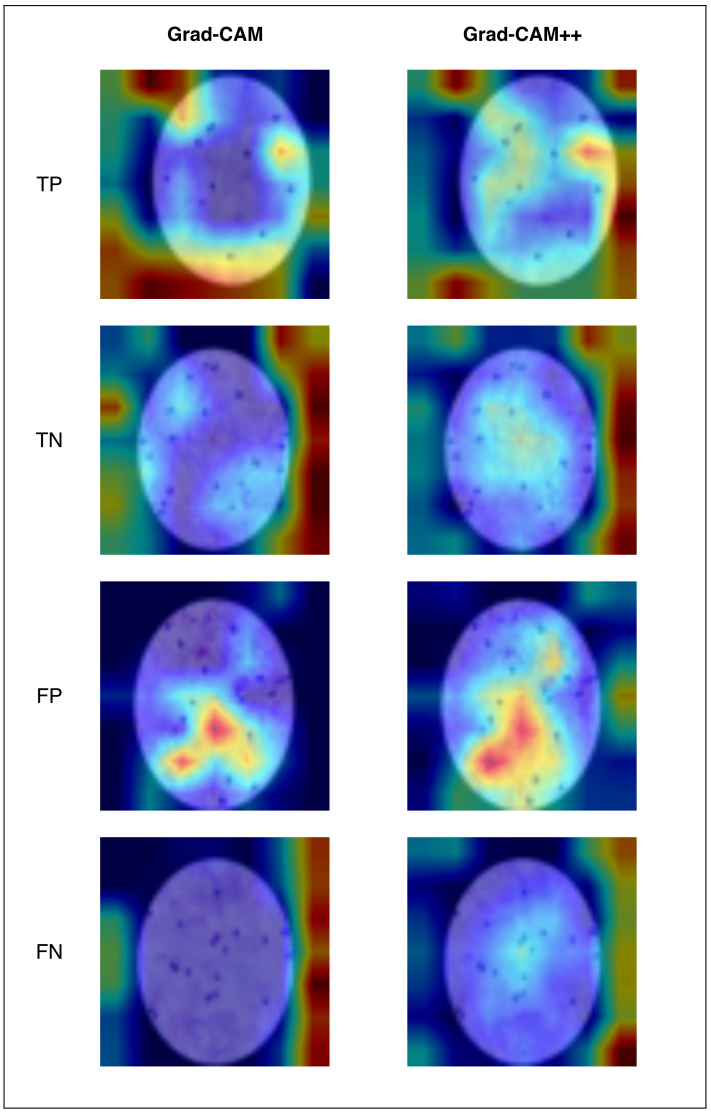



Figure 7.5: Comparison of Grad-CAM and Grad-CAM++ for MobileNetV3-Small LR3 predictions.

Based on these Grad-CAM and Grad-CAM++ results, all three architectures showed activation in some cases around the circular microscope boundary or the surrounding background. This finding does not prove that the models learned a shortcut by relying on edges, but it raises the question whether the image characteristics of the NLM dataset may have influenced some predictions. For this reason, an additional external test dataset with different image characteristics from the NLM dataset was used in the following section.

## 7.6 External Validation

To investigate whether the models may have relied on edge characteristics in the NLM dataset, an external test was performed on a new dataset that the models had never seen before. This dataset is from AIR Uganda. It contains 1182 malaria-positive images and has different image characteristics compared with the NLM dataset. No uninfected images are involved in the test.

The Uganda images were only used for external testing with already trained checkpoints. No additional training or optimisation was performed on the Uganda images.

Figure 7.6 shows one example image from the NLM dataset and one example from the AIR Uganda dataset to illustrate the difference in image appearance between the two datasets.

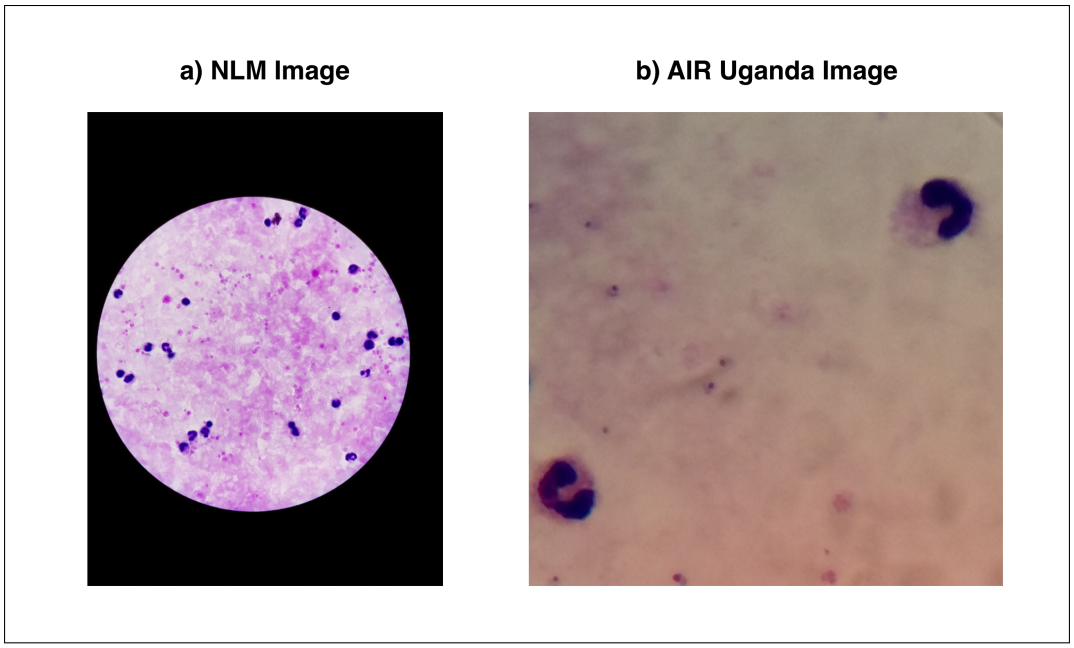



Figure 7.6: Example infected images from the NLM dataset and the AIR Uganda external dataset: (a) NLM (image TF1N_1522) and (b) AIR Uganda (plasmodium-phone-0001.jpg).

As mentioned before, the external dataset contains only infected images. For this reason, sensitivity was used as the main evaluation metric. Specificity could not be calculated because no uninfected samples were available. The results are shown in Table 7.6.

Table 7.6: External test results on the AIR Uganda dataset.

| **Model** | **Correctly predicted** | **False Negatives** | **Sensitivity** |
|---|---|---|---|
| ResNet50 HR | 1182 / 1182 | 0 | 1.0000 |
| ResNet50 LR3 | 1164 / 1182 | 18 | 0.9848 |
| ResNet18 LR3 | 973 / 1182 | 209 | 0.8232 |
| MobileNetV3-Small LR3 Opt. 2 | 1130 / 1182 | 52 | 0.9560 |

ResNet50 at full resolution detected all 1182 infected images correctly and reached a sensitivity of 1.0000. ResNet50 at LR3 also performed very well on the external dataset. It reached a sensitivity of 0.9848 and missed only 18 infected images.

The optimised MobileNetV3-Small at LR3 reached a sensitivity of 0.9560. It predicted correctly 1130 infected images and missed 52 cases. It is also worth mentioning that, despite the class weighting towards the uninfected class, the model still maintained a high sensitivity.

ResNet18 at LR3 showed the lowest external performance, with a sensitivity of 0.8232 and 209 false negatives.

These results show that the models were still able to detect infected images from an unseen dataset, but the performance was not equally stable across all architectures. ResNet50 showed the strongest external result at both HR and LR3, while ResNet18 showed the lowest external sensitivity.

The findings in the external test indicate that the activation around image boundaries observed in the Grad-CAM analysis does not automatically mean that the models fail on images with

different characteristics. Since this experiment tests the models only on positive samples, nothing can be said about specificity or how well the models predict uninfected images.

## 7.7 Discussion and Limitations

After evaluating all results, it is clear that all three architectures performed very similarly at full resolution (HR). The achieved accuracies from the models were around 0.996 and the differences between them were based on only a few wrong predictions. The effect of reducing the resolution became more visible at LR2 and significantly at LR3. For ResNet50, sensitivity remained high across all resolution levels, while specificity decreased sharply when the resolution decreased. To be more precise, specificity changed from 0.9901 at HR to 0.9852 at LR1. At LR2 it decreased even more to 0.9409. At LR3, it reached its lowest 0.7389. These observed metrics show that the reduced resolution caused a number of uninfected images to be misclassified as infected.

At LR3, the differences between the architectures became more clear. ResNet50 reached an accuracy of 0.9286, while the baseline MobileNetV3-Small reached 0.8930. After the two optimisation stages, MobileNetV3-Small did improve in terms of accuracy, which reached 0.9276. This optimisation made the MobileNetV3-Small performance very close to ResNet50, which further supports the argument for selecting MobileNetV3-Small as the lightweight architecture to be considered for future deployment.

This MobileNet optimisation also showed a clear trade-off between sensitivity and specificity. The first optimisation improved the sensitivity by reaching 0.9920, but specificity was only 0.6700. The second optimisation stage compromised slightly on sensitivity (0.9640) to increase specificity to 0.7931. This represents a more balanced result between the two classes.

For a screening application, which is the end goal of this overall project, a high sensitivity is very important to prevent a high number of false-negative case. If the deployment relies also on a second test that validates the model prediction, then the first optimisation stage could therefore also be considered despite its lower specificity. Stage 2 was selected by comparing the test-set metrics, so its reported performance may be slightly optimistic. The Uganda dataset was not used for this selection.

The Grad-CAM and Grad-CAM++ visualisations indicated that the models may not always focus only on regions inside the microscope field, but also on the edges. In several Grad-CAM outputs, activation was visible in the surrounding background. This raised the question whether dataset-specific image characteristics may have played a role in some predictions. This issue was investigated with a test using an external dataset from AIR Uganda, which showed that almost all models still score very well on images with different characteristics than NLM images. In the external test, ResNet50 reached a sensitivity of 1.0000 at HR and 0.9848 at LR3. The optimised MobileNetV3-Small also reached a high sensitivity of 0.9560. On the other hand, ResNet18 at LR3 performed badly and reached a sensitivity of only 0.8232. This shows that robustness was not the same for all architectures.

The strong external results suggest that the models did not rely only on the NLM image borders for their predictions.

The LR3 experiments are very relevant for the ocular adapter part of the overall project. These results show that effective classification was still possible even after a strong resolution reduction. This is considered positive for a future system where image quality may be low due to limited resources. Yet, LR3 was created by downsampling and does not represent real images captured directly through the actual ocular adapter, which means a real field performance cannot be determined here.

The external Uganda dataset also has some limitations. It contains only malaria-positive images. That said, specificity and false-positive behaviour could not be evaluated.

It's also worth mentioning that not all trained models were evaluated on the external Uganda dataset. ResNet18-HR, MobileNetV3-Small-HR, ResNet50-LR1, and ResNet50-LR2 were not included in the external test. In addition, the system developed in this thesis has not yet been evaluated on real field images from Ethiopia or under clinical conditions. These points should be investigated in future work.

# 8 Benchmarking

To have a perspective on where the performance of the developed models stands in the research field, the models are benchmarked against existing published studies that use thick blood smears.

The comparison considers relevant aspects such as the data used, whole-image versus patch-based approaches, and the reported performance metrics. Because the selected studies differ in these aspects, the comparison is indicative and not a ranking of the different methods. Smartphone-based and lightweight approaches are also considered, since they are relevant to the future deployment direction of this project and to the reduced-resolution experiments.

In the following sections, the selected studies will first be presented and then compared with the results of this thesis. At the end of the chapter, the main differences and findings will be discussed.

## 8.1 Selected previous Studies for Comparison

The selection of the studies for the benchmarking was based on whether they used thick blood smears, deep learning methods, and low-resource or smartphone-based settings. Lightweight models for resource-limited deployment were also given priority.

Two particularly relevant studies are Kassim et al. [22], published in 2021, and Yang et al. [8], published in 2020.

Kassim et al. [22] proposed PlasmodiumVF-Net for malaria diagnosis from thick blood smear images. Their system combines parasite detection with classification and was evaluated on a dataset containing P. falciparum, P. vivax, and uninfected samples. This study was selected because it uses the same NLM/NIH dataset family as the dataset used in this thesis and therefore provides one of the closest comparisons.

Yang et al. [8] developed a smartphone-based deep learning system for parasite detection in thick blood smear images. Their approach combines candidate screening with a CNN and was designed for use on mobile devices. This study is relevant because it combines thick smear microscopy, smartphone image acquisition, and deep learning.

Two other studies were also selected because they share similar aspects with the work in this thesis. These are Maturana et al. [17] and Vásquez Ascate et al. [24].

Maturana et al. [17] presented the iMAGING system, which combines a low-cost robotized microscope, smartphone image acquisition, and CNN-based parasite detection. Several object detection architectures were compared on thick blood smear images. This study was selected mainly because of its system-level similarity to the future deployment direction of this project.

Vásquez Ascate et al. [24] developed a smartphone-based system that detects parasites in thick blood smears using several lightweight object detection models. The dataset was collected from real clinical samples using different smartphone models and included both P. vivax and P. falciparum. This study was selected because it combines thick blood smears, smartphone acquisition, lightweight architectures, and deployment-oriented evaluation.

## 8.2 Comparison with Previous Studies

Table 8.1 summarises the results of this thesis together with the four selected studies. The comparison includes the data used, the architectures, the task, and the main reported results.

Table 8.1: Comparison with selected previous studies.

| Study | Data | Architecture | Task | Main Results |
|---|---|---|---|---|
| This thesis | 6038 NLM thick-smear images | ResNet50, ResNet18, MobileNetV3-Small | Whole-image binary classification | Best HR accuracy: 99.69%, MobileNetV3-Small at LR3: 92.76% |
| Kassim et al. [22] | Approx. 6000 images, 350 patients | Mask R-CNN and ResNet50 | Detection and image/patient-level classification | 83.9% image-level, 92.3% patient-level accuracy |
| Yang et al. [8] | 1819 images, 150 patients | Custom CNN | Patch classification: parasite vs. background | 93.46% accuracy |
| Maturana et al. [17] | 2571 thick-smear images | YOLOv5x and other detectors | Parasite detection | YOLOv5x: F-score 92.79% |
| Vásquez Ascate et al. [24] | 8625 images, 400 slides | YOLOv11n and other lightweight detectors | Parasite object detection | YOLOv11n: 90.92% accuracy, F1 93.8% |

The study of Kassim et al. [22] provides the closest comparison to this thesis because they use the same NLM/NIH thick-smear dataset family that was used in this work. On top of that, their method also includes ResNet50, which was one of the high-performing architectures during the testing phase. The main difference is that their PlasmodiumVF-Net first detects parasite candidates using Mask R-CNN and then uses ResNet50 to filter false-positive candidates before performing image- and patient-level classification. Their framework reached an overall accuracy of 83.9% at image level and 92.3% at patient level. On the other hand, the approach in this thesis directly classifies the complete image as infected or uninfected, with the best full-resolution models reaching 99.69% image-level accuracy. The higher accuracy in this thesis should be interpreted with care because Kassim et al. distinguish between P. vivax, P. falciparum, and uninfected samples, while this thesis uses a binary infected/uninfected classification.

The study of Yang et al. [8] also provides a relevant comparison because they use thick blood smear images for smartphone-based malaria detection. Their method first identifies potential parasites using an intensity-based Iterative Global Minimum Screening (IGMS). Then, it uses a customised CNN to classify each candidate as parasite or background. Their CNN reached an accuracy of 93.46%, a sensitivity of 92.59% and a specificity of 94.33%. The deployment-relevant model in this work is MobileNetV3-Small, which provides a useful comparison because Yang et al. target smartphone-based detection and MobileNetV3-Small was selected as the lightweight architecture for the future deployment direction of this thesis. At full resolution, MobileNetV3-Small reached an accuracy of 99.69%. After the optimisation at LR3, it reached an accuracy of 92.76%, a sensitivity of 96.40% and a specificity of 79.31%. The LR3 accuracy is close to the 93.46% accuracy reported by Yang et al., although the two approaches solve different tasks. Yang et al. classify regions as parasite or background, while MobileNetV3-Small in this thesis classifies complete images as either infected or uninfected.

The studies by Maturana et al. [17] and Vásquez Ascate et al. [24] are more focused on parasite detection and practical deployment. Both use object detection models on thick blood smear images and include smartphone-based image acquisition. Maturana et al. used YOLOv5x to detect leukocytes and Plasmodium trophozoites and reached an overall F-score of 92.79% for this detection task. Vásquez Ascate et al. used YOLOv11n for parasite detection and reached an accuracy of 90.92% and an F1-score of 93.8%. MobileNetV3-Small at LR3 in this thesis reached an accuracy of 92.76% and an F1-score of 95.45% for the binary classification of complete images

as infected or uninfected. The results are quite close in terms of the reported performance metrics. However, these metrics should not be understood in the same way because the models are solving different tasks. Maturana et al. and Vásquez Ascate et al. focus on detecting parasites inside the image, while the model in this thesis classifies the complete image as infected or uninfected.

## 8.3 Discussion of the Results

The models developed in this work show in general a high performance, especially with full-resolution images, with the best accuracy reaching 99.69%. This performance is higher than the results reported in all selected studies. However, this does not necessarily mean that the other approaches are inferior, since the tasks and their complexity are different. Binary classification is a simpler task compared with parasite detection or species classification. Even so, the performance achieved during the testing phase still shows that the developed models perform very well on the used dataset.

MobileNetV3-Small, which is the deployment-relevant model in this work, also reached good performance. After the optimisation at LR3, it achieved an accuracy of 92.76% and a sensitivity of 96.40%, although the specificity was 79.31%. This lower specificity may be related to the class imbalance in the dataset, since only 50 uninfected patients were available compared with 300 infected patients. The overall performance is also in a similar range to the deployment-oriented studies discussed in the previous section. This supports the idea that a lightweight model can still maintain good performance when working with reduced image resolution.

As mentioned in the previous chapters, the goal of the overall project to which this work belongs is malaria prescreening. In this context, the achieved sensitivity of 96.40% at LR3 is positive because most infected images are still detected. On the other hand, the specificity of 79.31% means that more uninfected images are incorrectly flagged as infected. For a prescreening system, this may be more acceptable than missing infected images, but the number of false-positive predictions should still be improved.

What cannot be proved from this benchmarking is whether one of the compared approaches is generally better than the others. The studies use different datasets, tasks, models, acquisition methods and evaluation metrics. In addition, the reduced-resolution experiments in this thesis were simulated from the available images and were not performed using images captured with the final smartphone and microscope setup.

Overall, the benchmarking with the four selected studies gives a useful indication that the results of this thesis are promising for the malaria prescreening task and that the models can still achieve good performance at lower resolutions. However, an evaluation using images captured with a smartphone and microscope setup is still required to test the performance under real deployment conditions.

# 9 Conclusion

The work of this thesis focused on developing a deep learning model for malaria prescreening with limited-resource conditions in mind, including the future use of smartphone-based images. This thesis is part of a larger system that also includes the development of an ocular adapter, which is intended to be attached to a microscope and enable the capture of thick blood smear images using a smartphone. The model developed in this work is intended to classify complete images as infected or uninfected. An important requirement was therefore also the evaluation of lightweight architectures that could later be suitable for deployment.

To achieve the intended goal of this work, the NLM/NIH thick blood smear dataset was selected for training, validation and testing. A patient-based split of 70% training, 15% validation and 15% testing was used. The selected architectures were ResNet50 as the main high-performance reference model, ResNet18 as a smaller ResNet architecture, and MobileNetV3-Small as the deployment-relevant lightweight model. Since the different parts of the overall project were developed in parallel, the final image resolution of the ocular adapter was not yet available during this thesis. For this reason, different resolution levels were simulated, starting from HR at 224 pixels and reducing the image resolution to LR3 at 56 pixels before resizing back to the model input size. This allowed the behaviour of the models under increasingly reduced image resolution to be investigated. Grad-CAM and Grad-CAM++ were also used to analyse which image regions influenced the predictions. In addition to the NLM/NIH test data, an external test using the AIR Uganda dataset was performed to evaluate the models on images with different characteristics from the development dataset.

The testing results showed that all architectures achieved very high performance at full resolution. ResNet50 reached an accuracy of 99.58% at HR. Its performance remained high at LR1 and LR2, but a clearer decrease was observed at LR3, where the accuracy dropped to 92.86%. The specificity also decreased to 73.89%, while the sensitivity remained high at 98.00%. ResNet18 and MobileNetV3-Small both reached the highest full-resolution accuracy of 99.69% and a sensitivity of 100%, meaning that all infected images in the internal test set were correctly classified. Under stronger resolution reduction, the main decrease was seen especially in the classification of uninfected images, with more false-positive predictions being produced. MobileNetV3-Small was therefore further optimised at LR3 and finally reached an accuracy of 92.76%, a sensitivity of 96.40%, a specificity of 79.31% and an F1-score of 95.45%.

The Grad-CAM and Grad-CAM++ results raised more questions about the behaviour of the models. In several examples, strong activations were visible around the edges and dark background of the microscope field instead of only inside the blood smear region. This raised the possibility that the models may partly use image characteristics that are not directly related to the parasites. To further investigate this, the external AIR Uganda dataset was used because its images have different visual characteristics compared with the NLM/NIH dataset. Since this dataset contains only infected images, only the sensitivity could be evaluated. ResNet50 reached a sensitivity of 100% at HR and 98.48% at LR3, with only 18 infected images missed at LR3. MobileNetV3-Small reached a sensitivity of 95.60% at LR3, while ResNet18 showed a lower sensitivity of 82.32% with 209 false-negative predictions. These results support that the models are not relying only on the specific borders or background characteristics of the NLM/NIH images. However, shortcut learning cannot be completely excluded. The external test also cannot show whether the models tend to predict the infected class when uncertain, because no uninfected samples were available in the AIR Uganda dataset.

Overall, the whole-image classification approach is technically promising based on the results of this work and the comparison with previous studies. Even when the image resolution was strongly reduced, good performance was still achieved. MobileNetV3-Small was the most relevant architecture for future deployment and also showed good performance at both high and low resolution.

The work of this thesis should be understood as a research prototype for future development and is not yet clinically validated for real-life malaria prescreening. The next important step is to obtain images captured with the combination of the developed ocular adapter, microscope and smartphone. These images will provide the real image resolution and characteristics that can be expected during future deployment. The models can then be tested using real captured images instead of only simulated resolution reductions. Future testing should also include more external datasets, especially datasets containing uninfected images, since this remained an open question during the testing phase. This would allow the specificity and false-positive behaviour of the models to be evaluated more reliably. In addition, using more uninfected samples during future model development could potentially help reduce the number of uninfected images that are incorrectly classified as infected.

A possible future extension of the system could also be parasite detection. After the binary prescreening approach has been further evaluated and improved, the system could be extended to identify parasites inside the image and later also distinguish between different Plasmodium species.

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